MILITARY SPECIFICATION

ATTITUDE HEADING REFERENCE SYSTEM AN/ASN-73

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

SCOPE

1.1 Scope - The equipment covered by this ecification shall measure aircraft angular attitude in tch and roll relative to the direction of gravity and in ading to an external reference. The AHRS is a component the Integrated Avionics Systems AN/ASQ-95(V), AN/ASQ-104, i AN/ASQ-105.

1.2 Classification - The equipment covered this specification shall consist of the following items:

Items	Type Designation	Appl. Para.
splacement Gyroscope	CN-105A/A	3.5.1
plifier - Power Supply	AM-4274/ASN-50	3.5.2
mpass Adapter - mpensator	MX-2826A/AJB	3.5.3
mpass Controller	C-2984/AJB-3A	3.5.4
te Switching roscope	CN-495A/AJB-3	3.5.5

1.3 Associated Equipment - This equipment all operate with the associated equipment listed in 6.5.

APPLICABLE DOCUMENTS

2.1 General - The following documents of e issue in effect on the date of invitation for bids form part of this specification to the extent specified herein:

FSC 6610

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SPECIFICATIONS

Military

MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Plate, Plastic, Lighting
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-M-18012	Markings for Aircrew Station Displays Design and Configuration of
MIL-T-18303	Test Procedures, Preproduction and Acceptance, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-19756	Transmitter, Remote Compass, Thin Win Type ML-1 (Unstabilized)
MIL-S-23603	System Readiness/Maintainability, Avi Systems Design, General Specification for

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SPECIFICATIONS (Cont'd)

Military

MIL-G-23723	Gyroscope, Rate Switching CN-495/AJB-3

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Test Levels and Accept/Reject Criteria for Reliability of Non-Expendable Electronic Equipment
MS 3116	Connectors, Plug, Electric, Solder Type, Straight, Bayonnet Coupling
MS 17322	Meter, Time Totalizing Miniature, Digital 115 Volt 400 Cycle
MS 17323	Meter, Time Totalizing Miniature Dial 115 Volt 400 Cycle

NAVAL AIR SYSTEMS COMMAND

WR-98	Microelectronic Devices used in Avionics Equipment, Procedures for

2.1.1 Availability of Documents - When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 1051, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.

3. REQUIREMENTS

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3.1 Preproduction - This specification ma provisions for preproduction testing.

3.2 Parts and Materials - In the selectio of parts and materials, fulfillment of major design objectives shall be the prime consideratcn. In so doing the following shall govern:

(1) Microelectronic items shall conform to requirements specified in WR-98 except as modified here

(2) Other parts and materials requir ments shall conform to MIL-E-540

(3) Non-repairable subassemblies, as outlined in MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approv by the procuring activity, unles previously approved by the procu ing activity.

3.2.1 Non-Standard Parts and Materials Approval - Approval for the use of non-standard parts and materials not previously approved by the procuring activity shall be obtained as outlined in MIL-E-5400.

3.2.2 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall be described by a contractor procurement specification and must be approved by the procuring agency unless previously appr by the procuring agency. Microelectronic Modular Assemblie (Microelectronic Circuits) (MIC's) shall be contained with Maintenance Modules (for example, on printed circuit cards)

3.2.3 Microelectronic Processes - The micr electronic modular assemblies shall be constructed from mo lithic (single chip) circuits or from thin film circuits. In the selection between monolithic and thin film circuits the monolithic circuits shall receive primary consideratio Monolithic circuits are defined as being constructed using processes whereby active and passive elements are formed b

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he diffusion of various materials into an active substrate. hin film circuits are defined as being constructed using rocesses whereby active and passive elements are formed by hemical deposition of various materials onto an inactive ubstrate.

3.2.4 Modules, Maintenance - The electronic ortions of the equipment comprising the AN/ASN-73 shall be ivided into maintenance modules (see MIL-S-23603). Mainten- nce modules shall normally be considered repairable.

3.3 Detail Requirements - Attitude Heading eference System (AHRS) AN/ASN-73

3.3.1 Construction - The equipment shall con- orm with all applicable requirements of MIL-E-5400, in onstruction and workmanship, except as otherwise specified erein. The Attitude Heading Reference System shall include he performance of the functions of an all attitude flight eference system. A continuous signal output of the dis- lacement of the aircraft about the pitch, roll and yaw axes nall be provided. The primary functions of the components re as follows:

(1) Displacement Gyroscope Assembly - The Displacement Gyro shall develop electrical signals representative of aircraft attitude through 360° of pitch, roll and azimuth for the attitude heading reference system.

(2) Amplifier-Power Supply - The Amplifier-Power Supply shall provide servo control for the displacement gyro and shall provide controlled application of power and excitation voltages for gyro starting and erection functions.

(3) Compass Adapter-Compensator - The Compass Adapter-Compensator shall operate in conjunction with the Compass System Controller. These two units, with the directional

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gyro and a remote compass transmitter, type ML-1, develop gyro stabilized magnetic, direct reading magnetic, and free gyro heading data for the system. Me of compensation for extraneous magnetic effects peculiar to the aircraft shall be provided.

(4) Compass System Controller - The Compass System Controller shall provide the controls necessary f a directional gyromagnetic compass heading system. It shall operate in conjunction with the compass adapter and shall provi controlled selection of "FREE", "SLAVED," or "COMPASS" mode of compass system operation.

The components of the system shall be so constructed that no parts will work loose in service and the System will wi stand the normal shocks, vibrations and such other conditi as are incident to service shipping, storage, and installa tion without failure.

3.3.1.1 Total Weight - The total weight of the equipment, excluding cables, shall not exceed 51.5 pou Component weights are:

(1)	Displacement Gyroscope	26.5 lbs
(2)	Amplifier-Power-Supply	8.3 lbs
(3)	Compass Adapter-Compensator	13.2 lbs
(4)	Compass System Controller	2.0 lbs
(5)	Rate Switching Gyroscope	1.5 lbs

3.3.2 Reliability -

3.3.2.1 Operational Stability - The AN/ASN-7 shall operate within the specified tolerances, continuousl or intermittently for a period of at least 120 hours witho

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e necessity for readjustment of controls which are inaccesble to the operator during normal use.

3.3.2.2 Operating Life - The AN/ASN-73 equipnt shall have a total operating life of 5000 hours with asonable servicing and replacement of parts. Parts requirg scheduled replacement shall be specified by the Contractor less otherwise specified herein.

3.3.2.3 Reliability in Mean Time Between Failes (MTBF) - For the test required under 4.4.3 the mean perating) time between failures shall be 275 hours.

3.3.2.4 Time Totalizing Meter - The system shall ntain time totalizing meters in accordance with MIL-M- 7793. e reading of the meters shall not require unsealing of the mponents. The time totalizers shall be installed within e present envelope of the components. Meters shall be cluded in the following components:

Component	Type of Meter
Displacement Gyroscope	MS 17323
Amplifier-Power Supply	MS 17323
Compass Adapter-Compensator	MS 17323

3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The equipment all provide for the use of cables and connectors in accornce with MIL-E-5400.

3.3.3.2 Interconnection Cabling - The equipment all be capable of satisfactory operation using external ring in accordance with the applicable requirements of L-W-5088. The external wiring shall be unshielded, except at a minimum number of the individual wires may be shielded en demonstrated as necessary to meet interference control quirements and provided the assembly of the cable to its ugs may be easily accomplished. External cables and that rtion of the connectors attached to the cables shall not supplied as part of the equipment.

3.3.4 Control Panels - All rack or console unted control panels shall conform to the applicable quirements of MIL-C-6781.

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3.3.5 Interchangeability of Reordered Equipment - For reordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previou manufactured equipment, supplied or designated by the procuring activity.

3.3.6 Interference Control - The generatio of radio interference by the equipment and the vulnerabili of the equipment to radio interference shall be controlled within the limits of MIL-I-6181.

3.3.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient ($25^{\circ}C \pm 5^{\circ}C$)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power and voltage	115 ± 1.0/200 ± 2.0 VAC and 27.5 ± 0.5 VDC

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environment service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 2 equipment, except as modified herein.

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3.3.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves IV and II of Specification MIL-E-5400.

3.3.10.2 Altitude - The equipment comprising the AHRS shall operate satisfactorily up to 20,000 feet.

3.3.11 Warm-Up Time - The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed 3.5 minutes under standard conditions and 5.0 minutes at extreme service conditions.

3.3.12 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts:

(1) AC Power (Three-Phase) 115/200V Category "B", 400 VA

(2) DC Power 28V, Category "B", .67 Amp

(3) AC Power (single phase) 26V-60VA

(4) AC Power (single phase) 5V-3VA

3.3.12.1 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.3.13 Cases - Cases of the components shall conform to applicable figures of this specification. Case and mounting flanges shall be made of non-ferrous, low density metal of uniform texture. Cases of the gyroscope assemblies shall be hermetically sealed. Cases of other units shall provide mounting for sealed electronic modules where applicable. Cases shall be finished with a lusterless black material, Color No. 37038 of FED-STD-595.

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3.3.13.1 Mountings - The component units shall be designed for shelf or bracket mounting, except for the compass controller, which is panel mounted.

3.3.13.2 Hermetic Sealing - The design of hermetically sealed cases for the gyroscope units shall be suc that the internal mechanism may be removed from the case, replaced, and the case resealed. This shall be accomplishe without the use of special tools and fixtures unless they are approved by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the unit may be subjected and such that it will pass the case lead test.

3.3.13.3 Filling Medium - The filling medium for the gyroscope units shall be of at least 98 percent purity free of dust particles and containing not more than 0.006 milligrams of water vapor per liter (due point-65°C) at the filling pressure. The filling medium shall be 95 percent helium and 5 percent carbon dioxide. The absolute pressure of the filling medium in the case shall be approxi mately 1 atmosphere.

3.3.13.4 Leak Rate - The leak rate shall not cause more than 10 percent loss or contamination of the filling medium after 1000 hours at a pressure differential of approximately 1 atmosphere.

3.3.14 Wiring - The internal wiring of each system component shall be neat, color coded and accomplishe in such a matter that individual wires may be easily traced

3.3.15 Dielectric breakdown - The materials used and the design shall be such that the resistance betwe isolated circuits and between each circuit and the case or connector shell is at least 50 megohms when a potential of 500 VDC is applied for a period of 10 seconds.

3.4 Performance - The Attitude Heading Reference System shall provide gyroscopically stabilized ro pitch, and heading reference data for aircraft. These shal be synchro type signals (3-wire).

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3.4.1 Attitude - The system shall provide aircraft roll and pitch outputs which are referenced to gravity. The outputs shall be gyro stabilized.

3.4.2 Heading - The system shall provide aircraft heading outputs which are referenced to North heading. The heading output data shall be available from three modes of operation as selected by the pilot.

3.4.2.1 Free Mode - In the free mode of operation, heading data shall be derived from a free running directional gyroscope. Provisions shall be made in the system to correct the data for gyro drift caused by earth's rate and by gyro drift bias.

3.4.2.2 Slave Mode - In the slaved mode of operation, heading output data shall be slaved to earth's magnetic and shall be gyro stabilized. Earth's rate and gyro drift bias corrections shall be supplied in this mode.

3.4.2.3 Compass Mode - In the compass mode of operation, heading data from the magnetic compass is repeated in the output without gyro stabilization. This is an emergency mode of operation intended to provide heading data only in case of gyro failure.

3.4.3 Output Accuracies - The system, when connected and operated as a system at standard conditions, shall be capable of providing the statistical accuracies of output signals as indicated below.

Output	Accuracy
Roll and Pitch	0.10 degree
Magnetic Heading	0.10 degree
System Heading Drift, DG Mode	0.25 degree per hour
Vertical Gyro Drift	5.0 degree per hour

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3.4.4 Performance at Environmental Extremes Performance deviations at environmental extremes shall be permitted as specified in the detail requirements.

3.5 Detail Requirements - AN/ASN-73 Subsystems -

3.5.1 Gyroscope Assembly, Displacement CN-1054A/A - The Displacement Gyroscope Assembly shall be designed and constructed to meet the requirements of this specification. No gimbal binding under differential pressure conditions shall occur as a result of interference between case and gimbal should a leak develop in service. It shall be designed to operate with the Amplifier-Power Supply

The Displacement Gyroscope Assembly shall consist of a vertical gyro and a directional gyro mounted in multiple gimbals with a common outer gimbal as described in the following paragraphs. This gimbal arrangement is used in order to avoid the gimbal lock position. The outline dimensions of the displacement gyro assembly shall be in accordance with Figure 1, pg. 13.

3.5.1.1 Vertical Gyro - The vertical gyro shall have sufficient angular momentum and be of such design so as to provide the performance specified herein when operated throughout the temperature, vibration, and acceleration ranges specified herein.

3.5.1.1.1 Gimbal System - The vertical gyro shall be suspended in a gimbal system having two free inner gimbals and a servo-driven outer gimbal to eliminate gimbal lock of the gyro. The outer gimbal shall be used in common with the directional gyro.

3.5.1.1.1.1 First Vertical Gyro Gimbal - The first gimbal shall be the gyro rotor housing. It shall be stabilized by the inertia of the gyro.

3.5.1.1.1.2 Second Vertical Gyro Gimbal - The second gimbal shall be the next gimbal outward from the first gimbal. It shall be stabilized in pitch by the gyro.

3.5.1.1.1.3 Third Vertical Gyro Gimbal - The third gimbal shall be the outer gimbal. It shall be servo-driven

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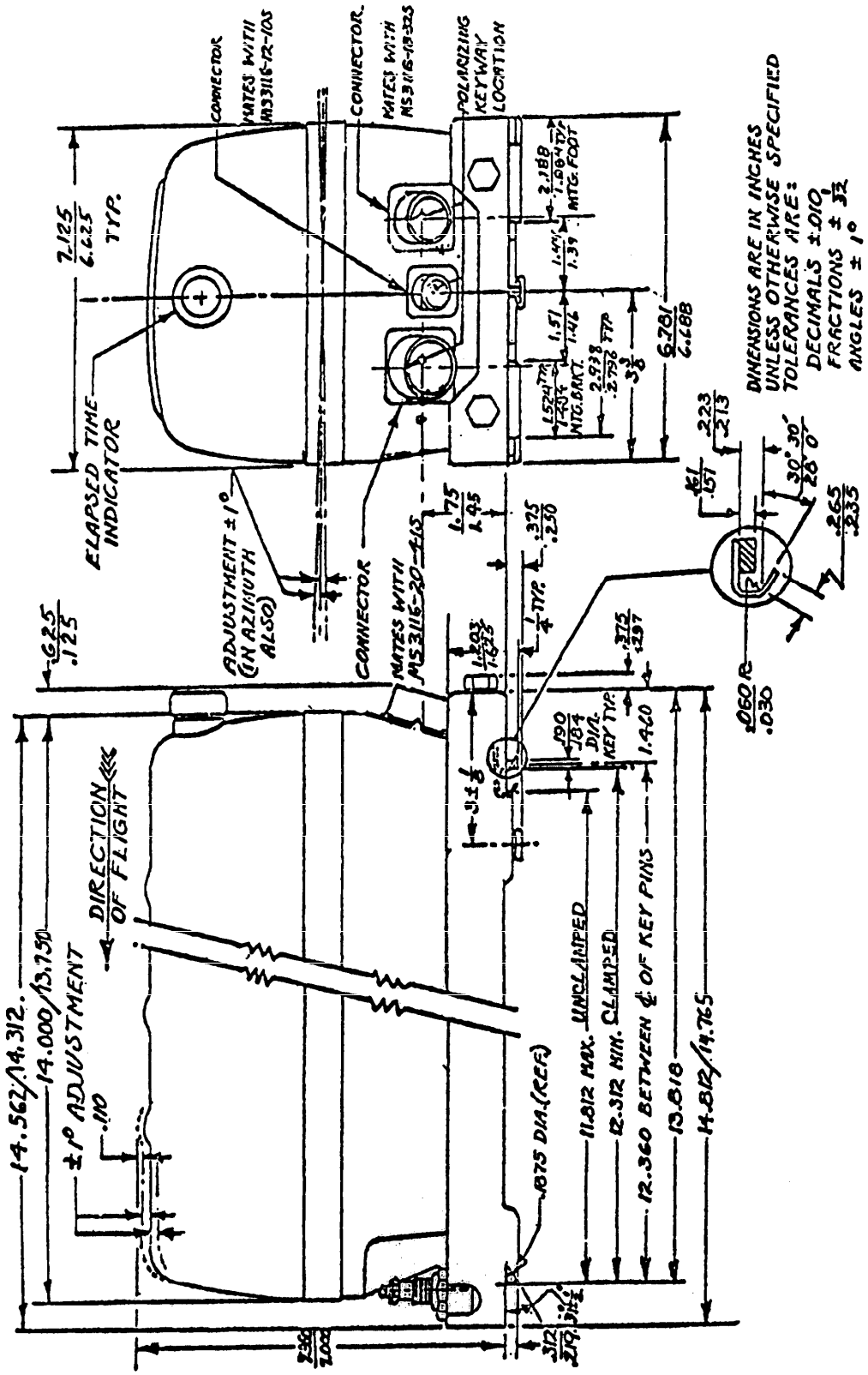


FIGURE 1 GYROSCOPE ASSEMBLY, DISPLACEMENT CN-1054A/A

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to maintain a fixed relation between the first and second gimbal except for certain conditions specified herein.

3.5.1.1.2 Vertical Gyro Axes of Freedom - The vertical gyro assembly shall have three axes of freedom. Each axis shall be a signal axis.

3.5.1.1.2.1 First Vertical Gyro Axis of Freedom - The first axis of freedom shall be between the first and second gimbal. This axis shall be maintained in the horizontal plane and be aligned with the orthogonal horizontal projection of the fore and aft axis of the reference assembl under normal conditions. A minimum freedom of $\pm 80^{\circ}$ shall be provided about this axis. An angular displacement about this axis shall provide an indication which changes from roll to yaw as the pitch angle changes from zero pitch to $\pm 90^{\circ}$ pitch. Stops shall be incorporated on this axis which produce a pitch precession when the freedom is exceeded and no damage shall result under this condition.

3.5.1.1.2.2 Second Vertical Gyro Axis of Freedom - The second axis of freedom shall be between the second gimbal and the third (servo-driven) gimbal. This axis shall be perpendicular to the first axis of freedom and shall be maintained in the horizontal plane under normal conditions. Unlimited freedom shall be provided about this axis. An angular displacement about this axis shall be a measurement of the pitch angle.

3.5.1.1.2.3 Third Vertical Gyro Axis of Freedom - The third axis of freedom shall be between the third gimbal (outer roll) and the instrument frame or case. This axis shall be perpendicular to the second axis of freedom and shall coincide with the fore and aft axis of the instrument. Unlimited freedom shall be provided about this axis. An angular displacement about this axis shall be a measurement of the roll angle.

3.5.1.1.3 Gimbal Actuation - The inner two gimbals are maintained in a fixed relation to the apparent grav vertical by the gyro erection system. The outer roll gimbal is actuated by a servo mechanism and is driven so as to main tain the spin axis of the vertical gyro and the pitch axis in a 90° relation to each other except for certain transient

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conditions. The response of the servo of the outer roll gimbal shall be such that this 90° relation will be maintained during roll rates from zero through 360° per second with minimum lag, and no overshooting, oscillation, hunting, or jumping of such a magnitude and frequency which will affect the performance of a high performance autopilot. The servo loop shall be tight enough so that during loop maneuvers when the outer roll gimbal is not controlled from external commands the outer gimbal rotates 180° in order to keep proper sensing of the servo drive when the outer roll gimbal axis passes through the vertical.

3.5.1.1.4 Erection - The erection mechanism shall incorporate separate systems for pitch and roll. Erection shall be accomplished by electrolytic switches and torque motors on the gimbals. When the vertical gyro is displaced in one axis, it shall not deviate in the other axis while it is erecting by more than 1.0°. The erection system shall erect the gyro from any attitude to the vertical as determined by the vertical sensing electrolytic switches. The pitch and roll erection torquer leads shall be brought out of the case so that erection cut-out may be provided and external voltages may be applied to process the gyro in pitch and roll. Proper erection voltages shall be obtained from the Amplifier-Power Supply to provide the following erection characteristics.

3.5.1.1.4.1 Initial Erection - With the proper input signals the vertical gyro shall have fast initial erection so that it assumes a position within 1° of vertical in pitch and roll within 1-1/2 minutes, or less, after power is applied.

3.5.1.1.4.2 Final Erection and Vertically - The vertical gyro shall align its spin axis with the direction of gravity within .25° in 2 minutes after the initial erection period. Five minutes after power is applied, the pitch synchro output shall not exceed 0.3 degree and the roll synchro output shall not exceed 0.4 degree from the direction of gravity.

3.5.1.1.4.3 Normal Erection Rate - The rate of normal erection about the roll axis shall be nominally 1.3° per minute and shall fall between the limits plotted on

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Figure 2 throughout all environments and roll attitudes of the control case. The rate of normal erection about the pitc axis shall be nominally 1.3° per minute and shall fall between the limits plotted on Figure 2 throughout all environ ments. Pitch erection sensing shall be maintained when displacing the gyro in roll ±45° or less.

3.5.1.1.4.4 Fast Erection - With the proper input signal, the gyro shall be capable of fast erection to the apparent vertical in both axis of the gyro simultaneously. The fast erection rate shall not be less than 15° per minute in either axis at room temperature or 12° per minute at temperature extremes.

3.5.1.1.4.5 Roll and Pitch Erection Cutout - The pitch and roll erection torquer leads shall be brought out of the case to provide a means to cut pitch erection, roll erection, or both, simultaneously. Pendulum erection cutout shall not affect the torquing of the gyro resulting from input of plane and earth rate correction.

3.5.1.1.4.6 Vertical Gyro Correction - Provision shall be made to apply torques in pitch and roll in either direction. The design shall be such that the torquers will work directly from either an amplifier or transformer. The torquers shall be internally excited such that when approximately 15 VAC, 400 cps, phase C+90° is applied to the control phase legs, the torquer shall cause the gyro to process at a nominal rate of 1° ± .2°/minute. The control phase impedance shall be 350 ± 70 ohms + J260 + 60 ohms. The power consumption in each torquer fixed phase shall not exceed 1 watt.

3.5.1.1.4.6.1 Earth Profile Correction - Provision shall be made within the gyroscope for earth profile corrections corresponding to Mach 0.6 or Mach 1.2. This shall be accomplished such that application of 26 VAC, phase C +90° shall torque the vertical gyro in pitch at a rate of .115 or .230 ± .05° per min. respectively. Interconnections shall be such that torque rates of 3.4.1.1.4.6 are not affected if no connection is made.

3.5.1.1.5 Vertical Gyro Synchro Outputs - Two

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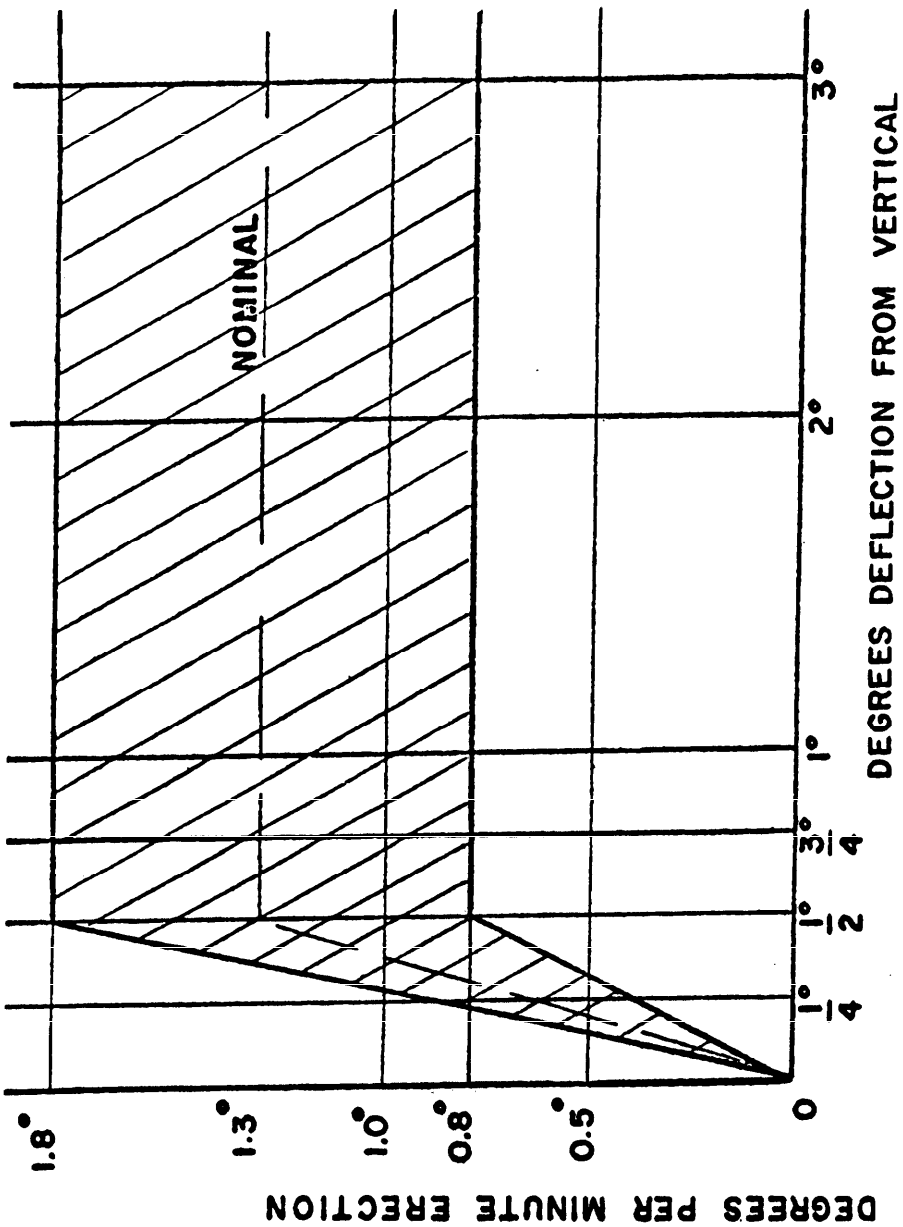


FIGURE 2 ERECTION RATE AS A FUNCTION OF DEFLECTION

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transmitting synchros shall be mounted on each vertical gyro axis of freedom except inner roll which shall have on These synchros shall provide displacement signals of gimbal movement about their respective axes at a one to one ratio.

One each of the synchros on the pitch and outer roll axes shall have their stator and rotor leads brought out to the signal electrical connector, Table I seperate from each other and other circuitry. These synchros shall each be capable of operating up to 8 high impedance control transformers each having an impedance of 220 $\pm$ 11 + J470 $\pm$ 24 ohms. The synchros shall have a nominal no load line-to-line output of 11.8 $\pm$ 0.25 V and phase shift of 4.0 $\pm$ 1.5^{c} When loaded with 8 CT's, the line-to-line output voltage shall not be less than 10.6V and the phase shift shall not be more than 10°.

The remaining synchros on the vertical gyro axes of freedc shall be internally excited with the voltage across terminals A and D of the control electrical connector, Table The stator leads of the synchros shall be brought out thrc the output connector terminals in accordance with Table II however, they may be connected to other control circuitry provided up to 6 high impedance control transformers havir characteristics as defined above can be operated external] from each. The internally excited pitch and outer roll synchro transmitters shall have a lead from the stator center tap brought out to connector terminals (Table III f use in failure detection circuits.)

3.5.1.1.5.1 Synchro Zeroing and Sensing - The transmitting synchros on the vertical gyro axes of freedo: shall be zeroed and their lead terminals of Table I connected to satisfy the procedures of 3.5.1.1.5.1.1 through 3.5.1.1.5.1.5 for the identification of the terminals. T angular spread between the nulls of the two synchros on t outer roll axes (and on the pitch axes) shall not exceed 0.4^{o}. The null voltages shall not exceed 30 millivolts.

3.5.1.1.5.1.1 Externally Excited Pitch Synchro - Terminals C and K Table I shall be connected to the power lead terminal A, Table II and terminal L of Table I shall be connected to the power lead terminal D, Table II. The

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TABLE I

Displacement Gyroscope Assembly Output Connector J361) Bendix P/N PT02H-18-32P, or equal, to mate with 33116P-18-32S or equal

Terminal	Function
A	Spare
B	Spare
C	Spare
D	Roll Switch - 90° - 270°
E H Rotor	Azimuth CX, ØC, TOP
F C Rotor	Azimuth CX, BOTTOM
G H Rotor	115 VAC excit. - F to G in phase with "X" to "Z" at zero displacement
H C Rotor	Roll CX - DG
J H Rotor	115 VAC excit. - H to J in phase with "Z" to "X" at zero roll
K C Rotor	Pitch CX - EXT.
L H Rotor	115 VAC excit. - K to L in phase with "Z" to "X" at zero pitch
M X Stator N Y Stator P Z Stator	Azimuth CX - BOTTOM
R X Stator S Y Stator T Z Stator	Azimuth CX - TOP
U X Stator V Y Stator W Z Stator	Roll CX - DG
X X Stator Y Y Stator Z Z Stator	Roll CX - VG (Internally excited rotor - neutral to ØC in phase with "Z" to "X" at zero roll
a X Stator b Y Stator	Pitch CX - EXT.
c Z Stator d X Stator	Pitch CX - INT

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TABLE I (Cont'd)

Displacement Gyroscope Assembly Output Connector (J361) Bendix P/N PT02H-18-32P, or equal, to mate with MS3116P-18-32A or equal

Terminal	Function
e Y Stator f Z Stator	(Internally excited rotor - ne tral to ØC in phase with "Z" to "X" at zero pitch)
g X Stator	Inner roll CX
h Y Stator j Z Stator	(Internally excited rotor - ne tral to ØC in phase with "Z" to "X" at zero displacement

TABLE II

Displacement Gyroscope Assembly Input Connector (J362) Bendix P/N PT02H-20-41P, or equal, to mate with MS3116P-20-41S or equal

Terminal	Function
A (Neutral)	Input Power - 3Ø 400 cps, 4 W
B (ØA)	200V line to line/115V line t
C (ØB)	neutral
D (ØC)	ØA, ØB, and ØC phase rotation
E	Spare
F	Leveling signal
G	Leveling excitation
H	Leveling torquer control Ø
J	Leveling torquer fixed Ø
K	Pitch torquer A
L	Pitch torquer B
M	Pitch switch A
N	Pitch switch B
P	Pitch switch common
R	Pitch capacitor

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TABLE II (Cont'd)

Displacement Gyroscope Assembly Input Connector (J362) Bendix P/N PT02H-20-41P, or equal, to mate with MS3116P-20-41S or equal

Terminal	Function
S	Power failure
T	Pitch Segment Switch
U	Slaving torquer fixed Ø
V	Slaving torquer control Ø
W	Pitch servo CT (H)
X	Reserved
Y	Reserved
Z	Pitch generator output (c)
a	Roll Switch B
b	Roll Switch A
c	Roll torquer excitation
d	Roll torquer B
e	Roll torquer A
f	Roll Generator output (hot)
g	Roll generator output (Cold)
h	Pitch generator output & pitch servo CT (Comm)
i	Leveling signal (A)
j	Leveling signal (B)
k	Reserved
m	Pitch M. G. control Ø
n	Pitch M. G. fixed Ø
p	Pitch generator excitation
q	Roll M. G. control Ø
r	Roll M. G. fixed Ø
s	Reserved
t	Roll generator excitation

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voltage across terminals A and B, and the voltage across terminals L and A or B of Table I shall be measured with the gimbals at zero displacement. The voltage across terminals A and B of Table I shall be at the minimum and the voltage across L and A or B of Table I shall be less than the voltage applied to terminals A and D of Table II. The voltage across terminals A and C and across B and C of Tabl I shall be measured as the reference assembly is rotated in climb about the pitch gimbal. The voltage across terminals A and C, Table I shall increase before decreasing the volta across the terminals B and C, Table I shall decrease before increasing.

TABLE III

Displacement Gyroscope Assembly Calibration Connector (J363) to mate with MS3116P-12-10s or equal	
Terminal	Function
A	Pitch torquer CT
B	Pitch torquer
C	Pitch torquer
D	Roll torquer CT
E	Roll torquer
F	Roll torquer
G	Roll Synchro Stator CT
H	Pitch Synchro Stator CT
J	Spare
K	Earth Profile Resistor

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3.5.1.1.5.1.2 Externally Excited Outer Roll Synchro - :minals W and H of Table I shall be connected to the ver lead terminal A of Table II, and terminal J of Table shall be connected to the power lead terminal D of Table . The voltage across terminals U and V of Table I and the .tage across terminals J and U or V of Table I shall be asured with the gimbal at zero displacement. The voltage oss terminals U and V of Table I shall be at the minimum l the voltage across J and U or V of Table I shall be ss than the voltage applied to terminal A and D of Table . The voltage across terminals U and W and across V and of Table I shall be measured as the reference assembly is ated about the outer roll gimbal in right roll. The vol- ge across terminals U and W shall increase before decreas- g and the voltage across the terminals V and W shall decrease fore increasing.

3.5.1.1.5.1.3 Inner Roll Synchro - Terminal J of ble I shall be connected to terminal A of Table II. The ltage across G and H of Table I and the voltage across rminal D of Table II and terminal G or H of Table I shall measured with the gimbals at zero displacement. The vol- ge across terminals G and H shall be at the minimum and e voltage across terminal D of Table II and terminals G H of Table I shall be less than the voltage applied to rminals A and D of Table II. The voltage across terminals and J and across H and J of Table I shall be measured as e first gimbal (inner roll) is held and the reference sembly with the outer roll gimbal is rotated in right ll. The voltage across terminals G and J of Table I shall crease before decreasing and the voltage across terminals and J shall decrease before increasing.

3.5.1.1.5.1.4 Internally Excited Pitch Synchro - rminal F of Table I shall be connected to terminal A of ble II. The voltage across D and E of Table I and the ltage across terminal D of Table II and terminal D or E Table I shall be measured with the gimbals at zero dis- acement. The voltage across terminals D and E shall be the minimum and the voltage across Terminal D of Table and terminals D or E of Table I shall be less than the ltage applied to terminals A and D of Table V. The ltage across terminals D and F and across E and F of Table shall be measured as the second gimbal (pitch) is held

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and the reference assembly with the outer roll gimbal is rotated in climb. The voltage across terminals D and F of Table I shall increase before decreasing and the voltage across terminals E and F shall decrease before increasing.

3.5.1.1.5.1.5 Internally Excited Outer Roll Synchr Terminal Z of Table I shall be connected to terminal A of Table II. The voltage across X and Y of Table I and the voltage across terminal D of Table II and terminal X or Y of Table I shall be measured with the gimbals at zero displacement. The voltage across terminals X and Y shall be at the minimum and the voltage across terminal D of Table I and terminal X or Y of Table I shall be less than the voltage applied to terminal A and D of Table II. The volt across terminals X and Z and across Y and Z of Table I sha be measured as the third gimbal (outer roll) is held and t reference assembly is rotated in right roll. The voltage across terminals X and Z of Table I shall increase before decreasing and the voltage across terminals Y and Z shall decrease before increasing.

3.5.1.1.6 Random Drift - With normal distribu the random drift of the vertical gyro shall not exceed 5° per hour - one sigma in the roll and pitch axes, respecti with the gyro spin axis at vertical.

3.5.1.1.6.1 Pendulosity Drift - To assure prope balance, the Vertical Gyro Assembly shall be so designed that the maximum free drift, noted after compensation for the earth's rotation, shall not be greater than 15° per h when operated over the entire temperature range specified herein, with the gyro spin axis tilted 20° of vertical.

3.5.1.1.7 Vertical Gyro Transmitting Synchros The externally-excited pitch synchro shall not exceed a maximum transmission error of four minutes of arc between $\pm$ 30 degrees. The angular spread between the nulls of th two synchros on the pitch axis shall not exceed 0.2 degre The null voltages shall not exceed 30 millivolts.

3.5.1.1.8 Power-off Freedom - When power is r moved momentarily, the vertical gyro shall still have fre about its first and second gimbal axes and shall not be

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displaced from the vertical during roll angles of ± 75° or pitch angles through ± 60° during this power-off condition.

3.5.1.1.9 Roll Segment Switch - A roll segment switch shall be incorporated on the third gimbal axis. This switch shall close terminal D of Table I to neutral from ± 90° through 180° to 270° of roll where zero roll is the initial zero position determined by 3.5.1.1.5.1.5. The circuit shall be open from the 270° roll position through zero roll to the ± 90° roll position.

3.5.1.2 Directional Gyro - The directional gyro shall be suspended in a gimbal system of which the outer gimbal is the third (servoed) gimbal of the vertical gyro. The directional gyro shall have three other gimbals suspended from this outer gimbal.

3.5.1.2.1 Inner Directional Gyro Gimbal - The first (inner) directional gyro gimbal (gyro wheel housing) shall be stabilized by the inertia of the gyro. Its pivot axis shall be the first directional gyro axis of freedom and shall be perpendicular to the spin axis of the gyro and shall normally lie in the horizontal plane. The slaving torquer shall act about this axis. This gimbal shall have a minimum of ±80° freedom about its axis.

3.5.1.2.1.1 Directional Gyro Azimuth Gimbal - The second directional gyro gimbal shall be stabilized by the inertia of the gyro, its pivot axis aligned with the vertical. The azimuth output synchros and the leveling torquer shall act about this axis. This gimbal shall be capable of continuous rotation about its axis.

3.5.1.2.1.2 Directional Gyro Pitch Gimbal - The third directional gyro (pitch) gimbal shall be servo driven so that it follows the vertical gyro second (pitch) gimbal at a maximum rate of not less than 200° per second. Its pivot axis shall be the third directional gyro axis of freedom and shall be parallel to the vertical gyro second axis (pitch) of freedom. This gimbal shall be capable of continuous rotation about its axis.

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3.5.1.2.2 Directional Gyro Leveling - The inner gimbal shall be slaved by means of the leveling torquer controlled by a suitable nonpendulous sensor so that the spin axis of the directional gyro is perpendicular to the azimuth axis. This leveling rate shall have an initial rate which will level the gyro within 1-1/2 minutes after power is applied. It shall have a normal leveling rate of 2° to 6° per minute.

3.5.1.2.3 Directional Gyro Slaving - The directional gyro shall contain an azimuth torquer to provide an azimuth slaving capability with a rate of 1° to 2° per minute with application of the proper signal.

3.5.1.2.4 Azimuth Transmitting Synchros - Two synchros shall be attached to the azimuth axis of the gyro to transmit signals proportional at a one to one ratio of the gyro. The synchros shall not exceed a maximum transmission error of 8 minutes of arc. The synchros shall be of the three phase wye connected stator and salient pole single phase rotor types. The angular spread between the nulls of the two azimuth output synchros shall not exceed 0.4°, and the null voltage of each synchro shall not exceed 30 millivolts. The azimuth transmitter synchros shall be capable of operating up to eight high impedance control transformer each having an impedance of 220 $\pm$ 11 ohms + J470 $\pm$ 24 ohms. The synchros shall have a nominal no load line-to-line output of 11.8 $\pm$ 0.25V and a phase shift of $4.0 \pm 1.5^{\circ}$. When loaded with 8 CT's the line-to-line output voltage shall not be less than 10.6V and the phase shift shall not be more than 10°.

3.5.1.2.4.1 Internally Excited Azimuth Transmitter The stator leads of the internally excited synchro shall be brought out to the terminals of the signal connector in accordance with Table I and shall be separate from other circuitry. The synchro shall be internally connected to neutral and shall be excited by the voltage applied to terminal E per Table I.

3.5.1.2.4.2 Externally Excited Azimuth Transmitter - The transmitter shall operate from a rotor excitatic of 115V, 400 cps and the maximum stator output voltage, line-to-line, shall be 11.8V. The transmitter shall have an electrical accuracy of $\pm$8 minutes or less. The rotor and

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ator leads shall be brought to separate pins of the elec-
ical connector per Table I, and shall be isolated from
her circuitry.

3.5.1.2.4.3 Synchro Zeroing and Sensing - Terminal
of Table I shall be connected to terminal A of Table II.
e voltage across R and S of Table I, and the voltage across
of Table II and R and S of Table I shall be measured with
e gimbals at zero displacement, the spin axis of the direc-
onal gyro aligned with the fore aft axis of the displacement
roscope assembly, and with the directional gyro spinning
(clockwise) when viewed from the aft end. The voltage across
rminals R and S of Table II shall be at the minimum and the
ltage across terminal D of Table I and terminal R or S of
ble I shall be greater than the voltage applied to terminal
and D of Table II. The voltage across terminals R and T
d across S and T of Table I shall be measured as the direc-
onal gyro second (azimuth) gimbal is held and the reference
sembly is rotated about its axis in right turn. The volt-
e across terminals R and T of Table I shall increase before
creasing and the voltage across terminals S and T of Table
shall decrease before increasing. When conducting this
ocedure pin E of Table I shall be connected to pin D of
ble II.

3.5.1.2.4.3.1 Externally Excited Azimuth Transmitter -
rminals P and F of Table I shall be connected to the power
ad terminal A of Table I and terminal G of Table I shall
connected to the power lead terminal D of Table II. The
ltage across terminals M and N of Table I and the voltage
ross terminals G and M or N of Table I shall be measured
th the gimbals at zero displacement, the spin axis of the
rectional gyro aligned with the fore-aft axis of the refer-
ce assembly and with the directional gyro spinning CW when
ewed from the aft end. The voltage across terminals M and
of Table I shall be at the minimum and the voltage across
rminal D of Table II and terminals M or N of Table I shall
greater than the voltage applied to terminals A and D of
ble II. The voltage across terminals M and P and across
and P of Table I shall be measured as the directional gyro
cond (azimuth) gimbal is held and the reference assembly
rotated about its azimuth axis in right turn. The volt-
e across terminals M and P of Table I shall increase before
creasing and the voltage across terminal N and P of Table
shall decrease before increasing.

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3.5.1.2.5 Directional Gyro Performance - The directional gyro shall be balanced so that the apparent drift due to earth's rotation is equal to the product of sine of the latitude of operation and 15° per hour in a CCW (counterclockwise) direction in the Northern Hemisphe and in a CW direction in the Southern Hemisphere.

3.5.1.2.6 Random Drift - With normal distribution the random drift of the directional gyro shall not exceed ±.20° per hour.

3.5.1.2.7 Compensatable Mean Drift - The mean drift of the directional gyro shall not exceed ±1.5° per hour from the computed apparent drift for the latitude of operation over the temperature range from -54°C to +71°C.

3.5.1.2.8 Pitch Segment Switch - A pitch segm switch shall be incorporated on the second gimbal axis (pitch servoed gimbal axis). This switch shall cause continuity to exist between pins T and A of J362 at all pitch gimbal angular displacements exceeding 90 ± 2° in either climb or dive from its normal level (zero) positic and shall provide open circuit at all displacements less than 90±2°.

3.5.1.2.9 Duration - After initial warm-up, the specified directional gyro performance shall be achie throughout a period of operation of not less than 2 hours

3.5.1.2.10 Directional Gyro Biasing - Whenever readjustment of directional gyro bias is made, a period c at least 2 hours during which all electrical power is dis nected shall intervene before initiation of testing to verify directional gyro drift performance.

3.5.1.2.11 Heading Error - After initial warm-up, the maximum instantaneous displacement of the heading synchro output from its initial value shall never exceed 1 degree for the duration specified in 3.5.1.2.9.

3.5.1.2.12 Mean Heading Error - The mean drift shall not exceed 0.22 degree at the end of the test peric specified in 3.5.1.2.9. Under temperature extremes the drift shall not exceed 0.5° per hour.

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3.5.1.3 Mounting - The mounting of the Displacement Gyroscope Assembly shall be in accordance with igure 1, pg. and such that the displacement reference ill not deflect by more than $\pm 1/4^{o}$ in pitch and roll or $.02^{o}$ in azimuth as a result of accelerations or any vibraion between 0 and 18 cps, with a minimum amplitude of 0.020 nch. The torsional resonant period shall be greater than 8 cps.

3.5.1.3.1 Direction Gyro Biasing - A means for ecording bias shall be affixed to the outer surface of the yroscope assembly.

3.5.1.3.2 Release - Means shall be provided or quick release of the reference assembly, including mount, rom the aircraft.

3.5.1.3.3 Adjustment - Means shall be provided or precision adjustment of the reference assembly in relaion to its mount.

3.5.1.4 Starting Time - The gyro rotors shall ttain full speed from rest within 5 minutes or less after pplication of power.

3.5.1.5 Slip Ring Continuity - Slip rings etween gimbals shall provide positive and continuous conact during aircraft maneuvers. Under the conditions listed elow, no sharp peaks or discontinuities shall occur in the synchro output signals when loaded with a servo test indicator having line-to-line impedance of 90 $\pm$ 15% + J570 $\pm$ 15% hms, or equivalent.

a. Rotating the reference case approximately 2 rpm in aximuth

b. Tilting the reference case approximately $\pm 80^{o}$ in pitch

c. Rotating the reference case approximately 2 rpm in roll, except that roll outputs shall also meet requirements at 1/2 rpm.

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3.5.1.6 Power Failure Signal - Loss of any phase of AC power shall cause a minimum of 15 volts to appe between the center-tap of the vertical gyro motor and powe neutral. The center-tap of the vertical gyro motor shall be connected to a terminal of J362 (see Table II, pg. 20).

3.5.1.7 Performance at Environmental Extreme When tested at environmental extremes, the displacement gy scope shall meet the erection, verticality, slaving, level ing, VG pendulosity, and DG mean drift requirements except as noted below.

3.5.1.7.1 Initial Erection - After one minute of operation the roll erection error shall not exceed 1.5°

3.5.1.7.2 Final Erection and Verticality - The final erection and verticality requirements shall be met within 5 minutes after the start cycle.

3.5.1.7.3 Output Signals - The output signals for roll, pitch, and azimuth shall not vary more than 1° during vibration error tests.

3.5.2 Amplifier-Power Supply AM-4274/ASN-5 The Amplifier-Power Supply shall be constructed to meet th requirements of this specification. The assembly shall operate with the gyroscope assembly. The outline dimensio and external characteristics shall be in accordance with Figure 3, pg. 31. The Amplifier-Power Supply unit shall fulfill the necessary servo and power requirements for pro per normal gyroscope operation. It shall contain the amplifiers for the servoed gimbals and related equipment and the starting and erection controls. Its connectors an terminals functions shall be in accordance with Tables IV and V.

3.5.2.1 Amplifier Channels - The amplifier channels shall be interchangeable, modular, plug-in-type where practicable. The amplifiers shall not incorporate electronic tubes. Parallel circuits shall be used where practicable so that partial failure does not result in total loss of operation.

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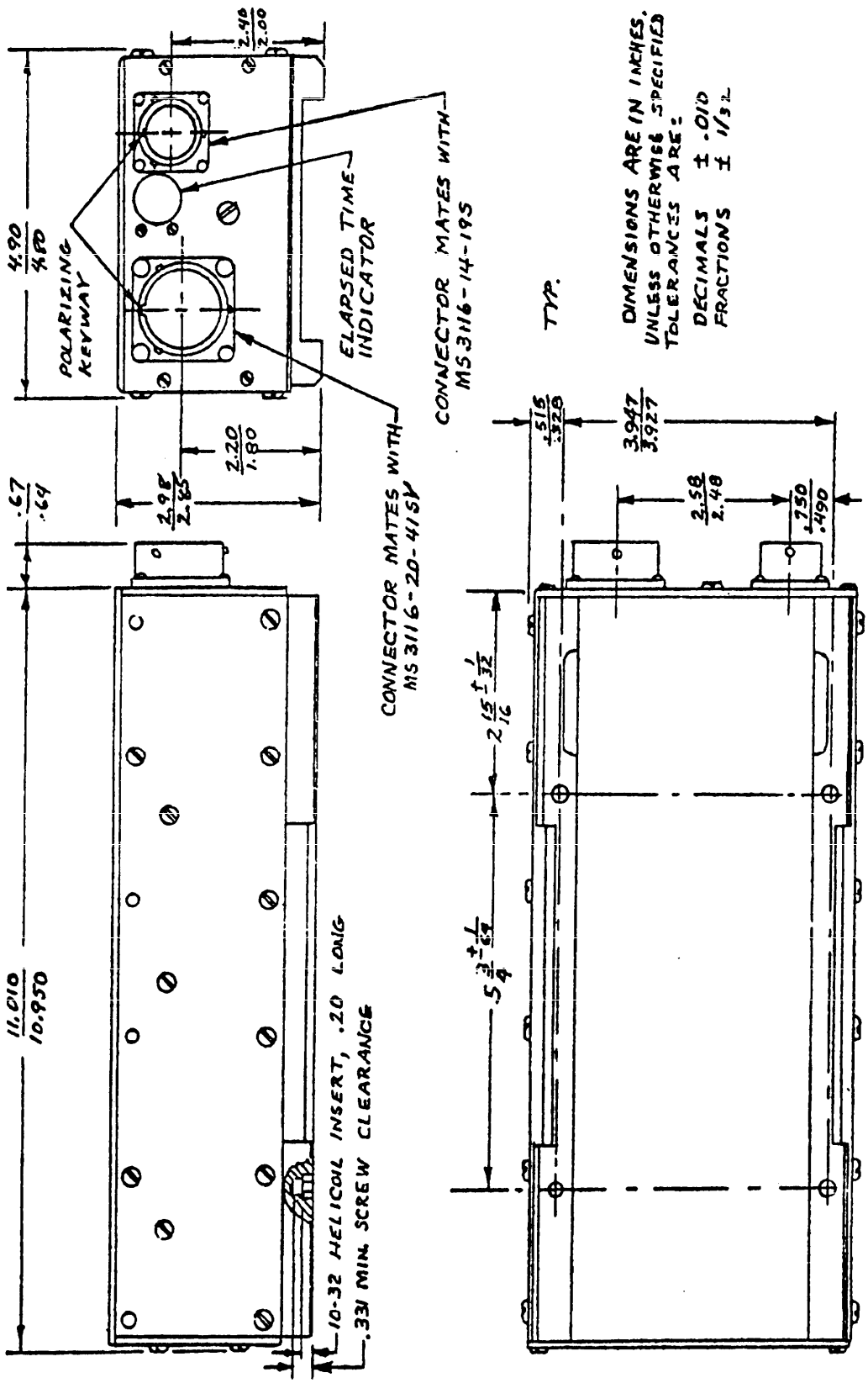


FIGURE 3 AMPLIFIER-POWER SUPPLY AM-4274/ASN-50

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TABLE IV

Amplifier-Power-Supply Input Connector (J202)
To mate with MS3116-14-19s or equal

Terminal	Function
A (Neutral)	115/200 VAC, 400 cps
B (ØA)	3 phase, 4 wire WYE
C (ØB)	Phase rotation A-B-C
D (ØC)	
E	Roll alarm - to F
F	Roll alarm - to E
G	Spare
H	Reserved
J	Reserved
K	Failure Warning
L	Failure Warning
M	Pitch erect cutout
N	Roll erect cutout
P	Cutout excit. - 115V, 400 cps, ØB
R	Start 60 sec. Auto sync - to S
S	Start 60 sec. Auto sync - to R
T	Reserved
U	Fast erection - to V
V	Fast erection - to U

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TABLE V

Amplifier-Power Supply Output Connector (J201)
to mate with MS3116-20-41SY or equal

Terminal	Function
A (Neutral)	Power 115/200 VAC 400 cps
B (ØA)	3 Phase 4 wire WYE Output to
C (ØB)	Phase rotation A-B-C Gyroscope
D (ØC)	
E	Test - Roll amp out
F	Leveling input
G	Leveling excitation
H	Leveling control phase
J	Leveling fixed phase
K	Disable alarm relay
L	Test - Roll driver out (amp in)
M	Test - Leveling modulator out (amp in)
N	Test - Alarm monitor
P	Pitch erection voltage
R	Pitch erection voltage
S	Test - Unfiltered DC
T	Test - Filtered DC
U	Pitch Alarm
V	Test - Pitch amp out
W	Pitch servo CT (H)
X	Roll M-G fixed phase (w/o third gimbal controller)
Y	Test - Leveling amp out
Z	Power failure
a	Roll servo input
b	Start cycle - 60 sec.
c	Roll erection voltage
d	Roll servo input (w/o third gimbal controller)
e	Roll servo input
f	Roll servo input
g	Start cycle - 60 sec.

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TABLE V (Cont'd)

Amplifier-Power Supply Output Connector (J201)
to mate with MS3116-20-41SY or equal

Terminal	Function
h	Pitch generator output (Signal ground)
i	Leveling input
j	Leveling input, common
k	Spare
m	Pitch M-G control phase
n	Pitch M-G fixed phase
p	Pitch generator excitation
q	Roll M-G control phase
r	Roll M-G fixed phase (w/third gimbal controller)
s	Test - Roll input transformer
t	Roll generator excitation

3.5.2.2 Starting Sequence Control - The sta: ing sequence control shall provide the signals to properl: erect and align the gyros. These signals are provided wh: the servoed gimbals are in any initial position and whene: the system power has been disconnected long enough to cau: recycling, even though the gyros may still be coasting. Proper alignment shall also occur when the servoed shafts and gimbals of the systems are rotated to any new positio: after power is disconnected.

3.5.2.2.1 0 to 12 Seconds - During the first : seconds after power turn on, the Amplifier-Power Supply s: provide reduced voltage for fast erection in pitch, open t power failure circuit, cage the outer roll gimbal, and pr: vide synchronizing signal to the compass adapter.

3.5.2.2.2 12 to 60 Seconds - During this peri: the Amplifier-Power Supply shall provide full voltage to gyro motors, provide leveling voltage, provide voltage fo:

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fast erection in roll and pitch, provide synchronizing signal to the Compass Adapter, open the power failure circuit, and cage the outer gimbal.

3.5.2.2.3 After 60 Seconds - After 60 seconds the Amplifier-Power Supply shall provide all necessary voltages for continuous operation of the gyroscope.

3.5.2.3 Erection Control - The erection control shall provide the functions of fast erection control, roll erection cutout, and pitch erection cutout. External leads shall be provided to permit Roll and Pitch erection cutout and fast erection voltage when commanded by external switches.

3.5.2.3.1 Fast Erection - Fast erection voltage of 50 volts, phase C, shall be provided during the first 60 seconds after power turn-on, or when commanded by external circuits.

3.5.2.3.2 Normal Erection - Unless fast erection is commanded the erection voltage shall be 15 volts.

3.5.2.3.3 Erection Cutout - Spearate control circuits be provided for cutting off roll and pitch erection voltages when commanded by external signals. Erection cutout from external signals shall be inoperative when fast erection is operated.

3.5.2.4 Servo Amplifiers - The Amplifier-Power Supply shall provide signals for Roll, Pitch, and Leveling, to the gyro.

3.5.2.4.1 Roll Servo Amplifier - The roll servo amplifier shall provide output to input voltages as follows:

Input Volts RMS	Output Volts RMS	Phase Relation
Input shorted	1 Max.	---
0.10 $\pm$ 0.002	21.$\pm$7	$180^{\circ} \pm 30^{\circ}$
0.35 $\pm$ 0.007	28 Min.	---

3.5.2.4.2 Pitch Servo Amplifier - The pitch servo amplifier shall provide output to input voltages as follows:

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Input Volts RMS	Output Volts RMS	Phase Relation
Input shorted	2.5 Max.	---
0.060 ± 0.002	10.0 ± 4	$0^{\circ} \pm 20^{\circ}$
0.350 ± 0.010	14.0 Min.	$0^{\circ} \pm 20^{\circ}$

3.5.2.4.3 Leveling Amplifier - The leveling amplifier shall provide output to input voltages as listed below: The output phase shall be either $0^{\circ} \pm 30^{\circ}$ or $180^{\circ} \pm 30^{\circ}$ with respect to ØC of input power depending on polarity of the input and required to properly torque the gyro.

Input Volts DC	Output Volts RMS
0.050 ± 0.002	10.5 ± 4
0.250 ± 0.005	20.0 Min.

3.5.2.5 Failure Warning - Failure warning signals shall be provided for Power, Roll Servo, and Pitch Servo Alarms. These signals shall provide same output as Power Failure Warning.

3.5.2.5.1 Power Failure Warning - A 3-phase warning circuit shall be incorporated in the amplifier. This circuit shall operate a set of normally open contacts which shall be separate from other circuitry. When proper power is applied to the control, the contacts will close after the initial erection cycle is completed. If at any time any phase of the 3-phase power to the control is broken, the contacts will open.

3.5.2.5.2 Servo Failure - The Amplifier-Power Supply shall have provisions for detecting an uncorrected err signal in the outer roll or pitch servo amplifiers, and for operating the power failure warning when an error is present. The failure alarm circuits shall be operative when externally connected to the amplifier inputs. An alarm disable relay shall be provided to disable the servo failure alarms when commanded by an external 28 VDC signal.

3.5.2.6 Power Interruption - Momentary interruption of power for periods less than 40 seconds shall not cause the starting circuits to recycle.

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3.5.2.7 Mounting - The mounting of the unit ill be such that the case can either be connected directly the airframe or permit a mounting plate and/or vibration olator to be inserted between the case and the mounting. inting dimensions are shown on Figure 3 pg. 31. When ojected to the high temperature test conditions of MIL-E-0, the unit shall be mounted on a heat sink capable of sipating 35 watts with a temperature differential of 3.5°C ween the unit and the chamber air.

3.5.2.8 Performance at Environmental Extremes - en tested at environmental extremes, the Amplifier-Power pply shall repeat within 20%, the results of its individual sts made at standard conditions prior to the start of vironmental tests, except that the servo amplifier maximum tputs shall be not less than 80% of specified outputs at andard conditions.

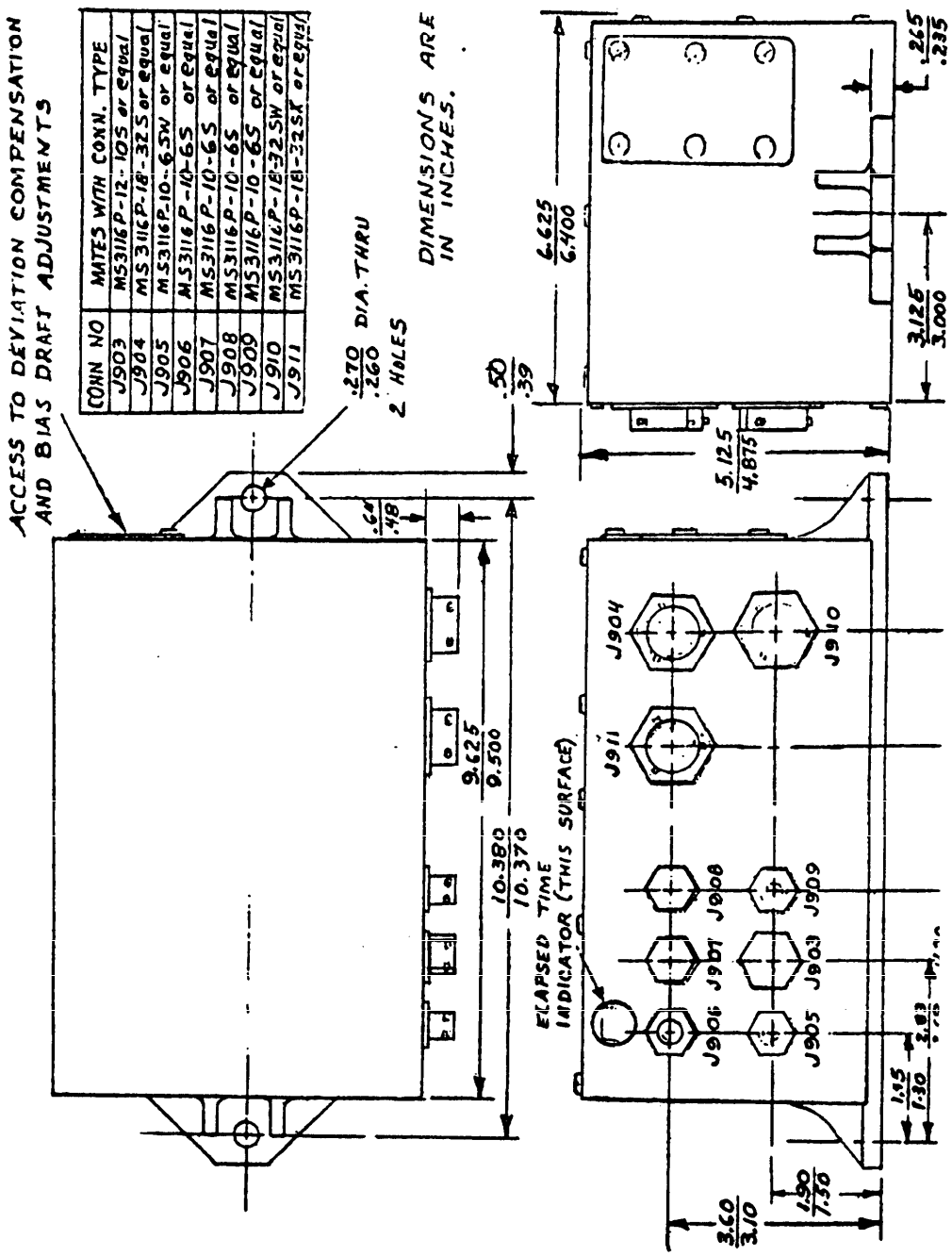
3.5.3 Adapter - Compensator Compass MX-2826A/B - The Compass Adapter shall furnish azimuth heading outts. It shall receive its basic inputs from the ML-1 Comss Transmitter, per MIL-T-19576, the directional gyro, and e compass controller. Dimensions shall be in accordance with gure 4, pg. 38. It shall provide the following functions: imuth (Directional Gyro) follow-up, magnetic slaving, heading synchronization, and DG corrections. Its connectors and rminal functions shall be in accordance with Tables VI through .

3.5.3.1 Azimuth Servo (DG Follow-Up) - The imuth follow-up shall repeat the directional gyro heading tput with high response and accuracy. This shall be direct tput from the Compass Adapter. The azimuth servo system all provide the following performance.

Static Accuracy	0° 15' RMS or 0° 30' Max.
Null Repeatability	0° 20'
Follow-up Rate, Minimum	180°/sec.
Follow-up Lag, Maximum	0.2°/RPM to 5 RPM
Natural Frequency, Undamped	5 CPS Min.

3.5.3.1.1 Dynamic Accuracy - The heading output ıall not change more than 0.25° when operating with the ıg. comp. oscillating up to ± 5° at 6 cpm.

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CONN NO	MATES WITH CONN. TYPE
J903	MS3116P-12-10S or equal
J904	MS3116P-18-32S or equal
J905	MS3116P-10-6SW or equal
J906	MS3116P-10-6S or equal
J907	MS3116P-10-6S or equal
J908	MS3116P-10-6S or equal
J909	MS3116P-10-6S or equal
J910	MS3116P-18-32SW or equal
J911	MS3116P-18-32SX or equal

FIG 4 ADAPTER, COMPENSATOR COMPASS MX-2826A/AJB

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TABLE VI

Compass Adapter to Controller Connector (J910) to mate with MS3116p-18-32SW or equal	
Terminal	Function
A	Latitude Compensation Supply
B	Latitude Compensation Supply
C	Compass Mode Relay
D	Slave Mode Relay
E	Latitude Compensation Signal
F	Latitude Compensation Signal
G	Neutral
H	Slew Motor Signal
J	Slew Pot Supply
K	Slew Pot Supply
L	Spare
M	Slew Relay
N	Spare
P	Sync, Indicator Signal
R	Sync, Indicator Signal
S	Sync Relay
T	Spare
U	Spare
V	Spare
W	Man. Slew Autopilot Decouple
X	Man. Slew Autopilot Decouple
Y	Spare
Z	Spare
a thru j	Spares

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TABLE VII

Compass Adapter Flux Valve Connector (J905) To mate with MS3116p-10-6SW or equal	
Terminal	Function
A	Compass CT input
B	Compass CT input
C	Compass CT input
D	Compass Transmitter excit.
E	Compass Transmitter excit.
F	Shield

TABLE VIII

Compass Adapter, CX Connectors (J906 thru J909) To mate with MS3116P-10-6S or equal (5 req'd)	
Terminal	Function
A	CX Output Y
B	CX Output Z
C	CX Output X
D	CX Excitation C
E	CX Excitation H
F	Spare

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TABLE IX

Compass Adapter Signal Connector (J904) To mate with MS3116P-18-32S or equal	
Terminal	Function
A	Erection cutout
B	Spare
C	Gyro CT input Y
D	Gyro CT input Z
E	Gyro CT input X
F	Azimuth CDX input X
G	Azimuth CDX input Z
H	Azimuth CDX input Y
J	Erection cutout
K	Erection cutout
L	Azimuth CDX output Y
M	Azimuth CDX output Z
N	Azimuth CDX output X
P	CX excitation source H
R	CX excitation source H
S	CX neutral source C
T	CX neutral source C
U	Gyro CT output
V	Gyro CT output
W	Azimuth amplifier input
X	Azimuth amplifier input
Y	Clutched CX excitation
Z	Clutched CX excitation
a	Clutched CX output Y
b	Clutched CX output Z
c	Clutched CX output X
d	Clutch
e	Clutch
f	Autopilot cutout
g	Turn cutout relay
h	Autopilot cutout
j	Turn cutout relay

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TABLE X

Compass Adapter Power Input Connector (J903) To mate with MS3116P-12-10S or equal	
Terminal	**Function**
A (Neutral)	Input power - 3Ø, 400 cps
B (ØA)	4 wire, 200V line-to-line/ 115V
C (ØA)	Line to neutral, ØA, ØB, ØC rotation
D (ØC)	
E	DC ground
F	+28 VDC
G	
H	Spares
J	
K	

TABLE XI

Compass Adapter Test Connector (J911) To mate with MS3116P-18-32SX or equal	
Terminal	**Function**
A	Tachometer Output
B	Tachometer Output
C	Spare
D	AGC Input
E	Reserved
F	Remodulated 400 cps
G, b	Slave and Correction Amplifier Output
H	Ground

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TABLE XI (Cont'd)

Compass Adapter Test Connector (J911) To mate with MS3116P-18-32SX or equal	
Terminal	Function
J	Pre-Amplifier-Limited Output, and Slaving and Correction Amplifier Input
K, e	Azimuth Amplifier Output
L	Azimuth Amplifier Input
M	800 cps Slave Preamp, Output
N	Common
P	AGC DC Control
R	Deviation Compensation
S	Earth's Rate Correction
T	Gyro Drift Correction
U	Slaving and Correction Motor
V	Common
W thru j	Spares

3.5.3.1.2 Reversing Azimuth Servo Null - Reversal the gyro CT output leads to the azimuth amplifier shall use a 180° rotation of the 4 and 20-power CX's but shall t reverse the differential synchro output.

3.5.3.2 Correction and Slaving Servo - The correction and slaving servo shall perform the functions of slaving the outputs of the compass adapter to the magnetic compass ansmitter and provide correction signals for the heading tputs corresponding to earth's rate (latitude) and directional ro drift bias.

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3.5.3.2.1 Magnetic Slaving - Magnetic slaving shall slave the output of the compass adapter to the magn compass transmitter. The slaving signals shall provide t following performance:

Static Accuracy	0° 30' Max.
Follow-up Rate	.7 to 1.5° per min.

3.5.3.2.2 Deviation Compensation - A 24 point electrical deviation compensator shall be provided to min mize errors of the remote compass transmitter and its ass iated control system. It shall only be effective when th magnetic compass signal is being utilized. The 24 point viation compensator shall provide compensation with minim range of $\pm$ 3.5 degrees at each adjustment point.

The deviation compensator shall be a plug-in module and b easily accessible. The Compass Adapter Compensator shall contain provisions which will permit elimination of the deviation compensation signal from the slaving signal as aid in magnetic compass compensation.

3.5.3.2.3 Automatic Gain Control - The Compas Adapter shall contain automatic gain control to maintain slaving rate requirements when supplied with signal from ML-1 magnetic compass operating in a magnetic field with horizontal component varying between 0.08 and 0.4 oersted

3.5.3.2.4 Heading Corrections - Signals shall be available in slave and free modes to compensate for th DG mean drift up to 5° per hour and for the apparent DG drift caused by earth's rotation up to 15° per hour. The correction servo shall integrate these signals and provi a controlled rate of correction under standard condition as follows:

Range	0.5° to 20° per hour min.
Random	$\pm$ 0.15° per hour rms.
Maximum error	$\pm$ 0.3° per hour

3.5.3.3 Synchronization - Means shall be provided to automatically synchronize the directional gy input signal to the magnetic compass transmitter at a fa rate during an initial power application, during switchi

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to slave mode, or by manually depressing the push-to sync button on the controller. Synchronization shall be at a minimum rate of 20°/sec. and shall be accurate to ± 2°.

3.5.3.3.1 Indicator Signal - A synchronization signal shall be provided for driving the Compass Controller synchronizing indicator meter. This signal shall provide a current of 10 ± 3 microampere per degree (1100 ohm load) and shall be limited to a maximum of 250 microamperes. Polarity of the signal shall indicate the direction of the error. The signal at null shall not exceed ±5 minroamperes.

3.5.3.4 Set Heading Control - The Compass Adapter in combination with the Compass Controller shall provide for setting of heading while in "FREE" mode of operation. The output heading data from the Compass Adapter-Compensator shall increase (+) when the knob of the Compass Controller is rotated clockwise. Speed of this rotation shall be adjustable from 1 to 9 RPM as determined by how far the set reading knob is rotated.

3.5.3.5 Data Transmitting Synchros - The outputs of the Compass Adapter shall be obtained from the following:

a. One 20-power transmitter
b. Three 4-power transmitters
c. One differential transmitter
d. One clutched synchro

Items a, b, d and e shall be driven by the azimuth servo, and item c shall be driven by the correction and slaving servo.

3.5.3.6 Clutched Synchro - The clutched synchro shall be excited thru a fixed resistor in series with the rotor. Synchro and resistor shall be such that, when the output from 2 stator leads (blue to blk or yel to blk) are loaded with 5000 ± 10% ohms and the synchro rotor (with resistor) are excited with 26 ± .5V, 400 ± 8 cps; the maximum output shall be 10.2 ± 1.2 volts at a phase angle of 30° ± 2° lead.

3.5.3.7 Synchro Zeroing - The output synchros shall be set at electrical zero for North (0°) heading output.

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The differential synchro shall be zeroed such that when the rotor is excited by a synchro transmitter operating at its electrical zero, the stator output will be at its electrical zero + 180°.

With the CX (J908) at 0° and the transmitter at 0° ± .5° and slaving cutout, the output of the CX's (J906, J907 and J909) shall be 0° ± 12'. The output of the CDX shall be 180° ± 30'.

NOTE: The heading synchro is adjusted for zero when the voltage from X to Y is at a minimum and the dial reading is 0° (NORTH), and with rotor lead C connected to stator lead Z the voltage from H to X shall be greater than the voltage frc H to C. The voltage between X and Z shall increase before decreasing for a rotation of the dial in increasing heading direction. For decreasing heading rotation of the dial, the voltage between X and Z shall decrease before increasing.

3.5.3.7.1 Clutched Synchro Zero - The output of the clutched synchro shall be 0 ± 1° with the clutch de-energized. With the clutch de-energized by 28 VDC, the output shall increase for increasing heading and decrease for decreasing heading change up to ±25°.

3.5.3.8 Compass Mode Operation - In the Compass Mode Operation, the output data transmitter follows the Magnetic Compass transmitter input. (No directional gyro stabilization). The compass mode shall be initiated by an external signal from the mode selector switch. Static accuracy in the compass mode shall be ± 1° 30'.

3.5.3.9 Performance at Environmental Extremes - When tested at environmental extremes, the Compass Adapter shall meet the positioning accuracy, positioning rate, and synchronizing requirements except as indicated below. Adjustment of deviation potentiometers shall not be required during environmental tests.

3.5.3.9.1 Azimuth Servo - The free mode positional tolerances shall be increased from ± 30' to ± 45' max, and dead zone from 20' to 30'.

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3.5.3.9.1.1 Compass Mode - The positional tolerances shall be increased from 1° 30' to 2°.

3.5.3.9.2 Magnetic Slaving - The rate shall be increased from 3.5 and 7.5 degrees in 5 minutes to 2.5 and 5.5 degrees in 5 minutes. The positional accuracy shall be increased from 30' to 1.0°.

3.5.3.9.3 Synchronization - The positional tolerances shall increase from ±2° to ±2.5°.

3.5.3.9.4 Synchronization Indicator - The current through the 1100 ohm resistor shall increase from 5 microamperes or less to 8 microamperes or less. The current through the 1100 ohm resistor for a 5 degree displacement of the magnetic compass transmitter shall increase from 10 ± 3 microamperes to 10 ± 4 microamperes.

3.5.3.9.5 Set Heading Control - The minimum speed of 9 RPM shall be decreased to 8 RPM.

3.5.3.9.6 Heading Corrections - The maximum error shall be ±0.5° per hour and the random error shall be ±0.20° per hour.

3.5.3.10 Excitation Voltage Output - The Compass Adapter shall provide the following excitation voltages:

(1) 26 V., 400 cps, for excitation of synchro transmitters. The output shall be capable of simultaneously exciting the transmitters of 3.5.3.5 at room temperature.

(2) 23.5 V, 400 cps, for excitation of the type MS-1 magnetic compass transmitter.

(3) 26 V., 400 cps, for excitation of the latitude compensation resolver in the compass controller.

(4) 3-0-3V. (6.0 V C T), 400 cps, for excitation of the heading set

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potentiometer in the Compass Controller.

3.5.3.11 Cutout Relay - A 28 VDC relay shall b provided which, when energized, will interrupt the slaving operation of 3.5.3.2.1. The relay shall also interrupt a gyro erection circuit routed through the Compass Adapter.

3.5.3.12 Autopilot Decoupling - A circuit rou through the Compass Adapter shall be interrupted whenever either manual or automatic synchronization occurs. The int ruption shall occur prior to the start of synchronization.

3.5.3.13 Mechanical Requirement - The servo-mechanism assembly shall be modular in construction and be of the plug-in type.

3.5.4 Controller-Compass System C-2984/AJB-3A - The Compass Controller shall be constructed to mee the requirements of this specification. Outline dimension shall be in accordance with MS17392. It shall operate in conjunction with the Compass Adapter defined by 3.5.3.

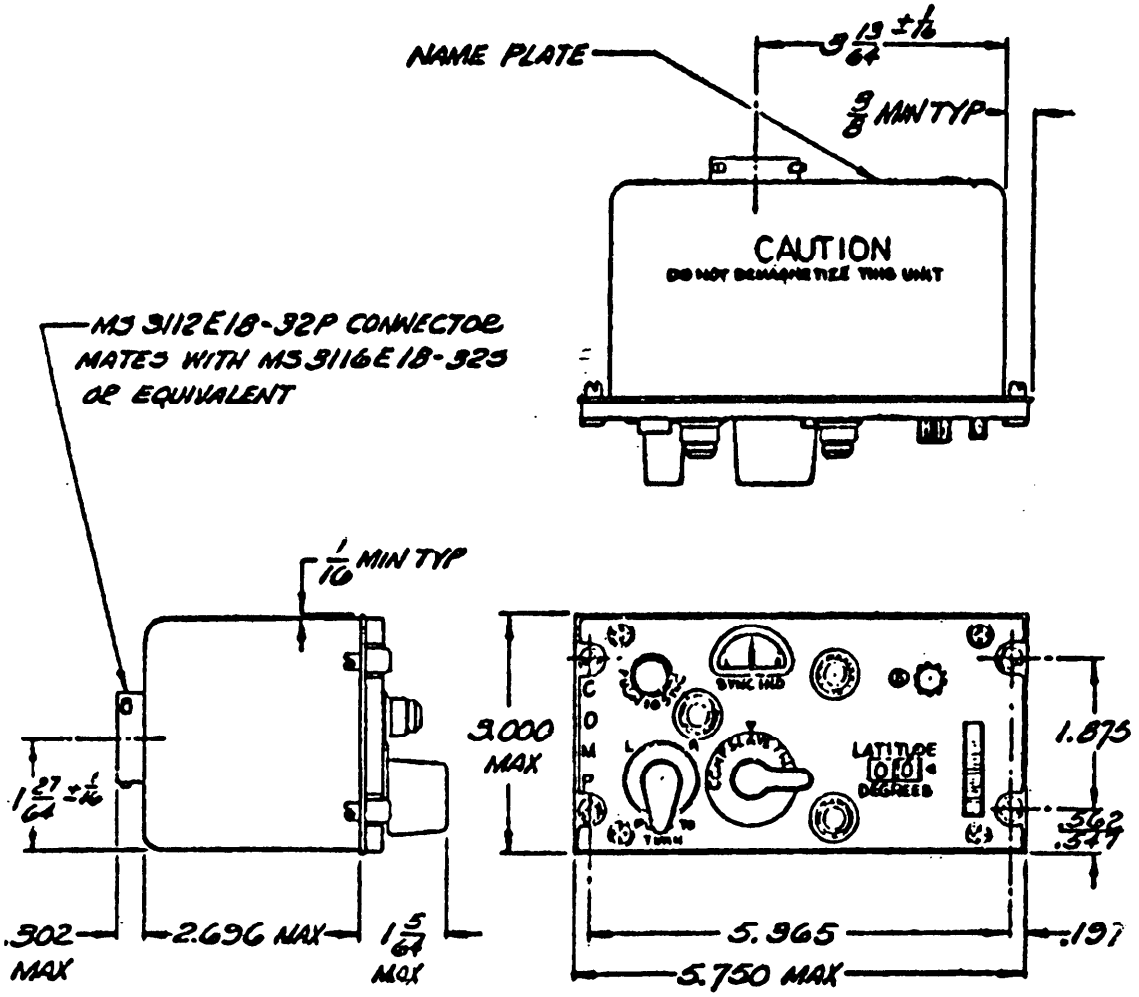
The Compass Controller shall provide the mode selection and controls for the operation of a directional gyro-magnetic compass heading system. The outline dimensions shall be in accordance with Figure 5, pg. 49.

3.5.4.1 Controls - The Compass Controller shall provide the following controls as shown on MS17392.

3.5.4.1.1 Mode Selector Switch - A switch shal be provided to select one of the following modes: "FREE", "SLAVED" and "COMPASS". The mode selector switch shall provide a circuit to operate compass and slave mode relays in the Compass Adapter. A second set of contacts shall be used to operate the synchronizing indicator in compass and slave modes.

3.5.4.1.2 Set Heading Control - The set headin control shall be labeled "PUSH TO TURN", shall provide swit ing to de-couple the autopilot and to control the directio and rate of slewing the system to an azimuth heading when operating in the "FREE" mode. The control shall consist o a Push-to-Turn switch and potentiometer. The switch shall,

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NOTES:

1. DIMENSIONS SHOWN ARE FOR INSTALLATION PURPOSES ONLY.
2. TOLERANCE ON DECIMAL DIMENSIONS SHALL BE ±.005 UNLESS OTHERWISE SPECIFIED.

FIGURE 5 CONTROLLER, COMPASS SYSTEM C-2984/AJB-3A

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when operated, provide one circuit to operate the slew circuits of the Compass Adapter, and a second circuit to open autopilot cutout.

3.5.4.1.2.1 Potentiometer - The set heading potentiometer shall have nominal resistance of 1000 ohms. The resistance between the fixed center tap and each end sh be equal within 80 ohms.

3.5.4.1.3 Synchronizing Switch - The synchroniz switch shall be labeled "PUSH TO SYNC" and shall provide fc synchronization of the azimuth heading output to the magnet compass when operating in the "SLAVED" mode.

3.5.4.1.4 Latitude Control - The latitude contr shall provide means to compensate for apparent drift of the gyroscope resulting from rotation of the earth. With 26 ± VAC applied and a load of 17.5K ohms ± 1%, the ratios betwe the in-phase component of the latitude control voltage and excitation voltage shall be as specified below for each lat tude setting:

LATTITUDE SETTING	VOLTAGE RATIO
90°	.777 ± 1%
75°	.750 ± 1%
60°	.672 ± 1%
45°	.549 ± 1%
30°	.388 ± 1%
15°	.201 ± 1%
0°	.005 (max)

For the "N" position of the hemisphere switch, the latitude control voltage shall lead the excitation voltage by 8° ± 3 and lag by 172° ± 3° for the "S" position of the hemisphere switch.

3.5.4.1.5 Hemisphere Switch - The hemisphere switch shall enable selection of the hemisphere in which la tude correction is to be applied.

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3.5.4.2 Control Knobs - The control knobs and marking thereon shall be in accordance with MS17392.

3.5.4.3 Synchronizing Indicator - The Compass Controller shall include a synchronizing indicator to provide a visual indication of synchronization between the azimuth heading output and the magnetic compass. With the mode selector in the "SLAVED" or "COMPASS" position, 45 microamperes DC shall cause a minimum pointer deflection of 2/3 of full scale from its center position. Deflection shall be to the left when a left turn is required to synchronize.

3.5.4.3.1 Dial Marking - The synchronizing indicator dial shall be marked as shown on MS17392. The "L" and "R" and zero index markings shall be lusterless black, Color #37038 of FED-STD-595 and the scale shall be lusterless white, Color #37875 of FED-STD-595.

3.5.4.4 Panel Illumination and Marking - The Compass Controller shall contain a lighting plate conforming to MIL-P-7788. The plate shall be marked in accordance with MIL-M-18012.

3.5.4.5 Mounting - Mounting provisions shall be in accordance with MIL-C-6781, and MS17392.

3.5.4.6 Connection - The electrical connector and terminal functions shall be in accordance with Table XII.

TABLE XII

Compass Controller Assembly Connector (J801) To mate with MS3116E-18-32S or equal	
Terminal	**Function**
A	Latitude compensation excitation
B	Latitude compensation excit.
C	"COMPASS" mode control signal
D	"SLAVE" mode control signal
E	Latitude compensation signal

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TABLE XII (Cont'd)

Compass Controller Assembly Connector (J801) To mate with MS3116E-18-32S or equal	
Terminal	Function
F	Latitude compensation signal
G	Neutral
H	Slew control signal
J	Slew control excitation
K	Slew control excitation
L	Spare
M	Slew mode control signal
N	Spare
P	Synchronizing indicator
R	Synchronizing indicator
S	Synchronizing switch
T	Lighting - 28V
U	Lighting - Ground
V	Spare
W	Autopilot Decouple
X	Autopilot Decouple
Y	Spare
Z	Spare
a	Spare
b	"FREE" mode control
c	"FREE" mode control
d	Spare
e	Cosine latitude signal
f	Cosine latitude signal
g	Spare
h	Spare
j	Spare

3.5.4.7 Performance at Temperature Extremes - The Compass Controller shall perform as specified herein at temperature extremes except as follows:

3.5.4.7.1 Voltage Ratio - The voltage ratio tolerances of ± 1% in 3.5.4.1.4 shall be ± 2%.

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3.5.4.7.2 Phase Angle - The phase angle toleran-s of $\pm 3^{\circ}$ in 3.5.4.1.4 shall be $\pm 5^{\circ}$.

3.5.4.7.3 Illumination - Illumination shall be ecked to note that lighting is present, but no quantitative asurements shall be required.

3.5.5 Gyroscope, Rate Swtiching CN-495A/AJB-3 - e rate switching gyroscope shall be in accordance with L-G-23723. The outline dimensions shall conform to Figure pg. 54 .

e gyroscope shall provide switching during turns. The roscope shall consist of a rate gyro and associated turn-itching circuits.

QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless herwise specified in the contract or purchase order, the pplier is responsible for the performance of all inspec-on requirements as specified herein. Except as otherwise ecified, the supplier may utilize his own facilities or y commercial laboratory acceptable to the Government. The vernment reserves the right to perform any of the inspec-ons set forth in the specification where such inspections e deemed necessary to assure supplies and services conform prescribed requirements.

4.1.1 Classification of Tests - The indivi-al equipments covered by this specification shall be subject-to the following tests to determine compliance with all plicable requirements:

(1) First Article Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Longevity Test

te: The term "First Article" has the same meaning as the term "Preproduction Model". The terms may be used interchangeably.

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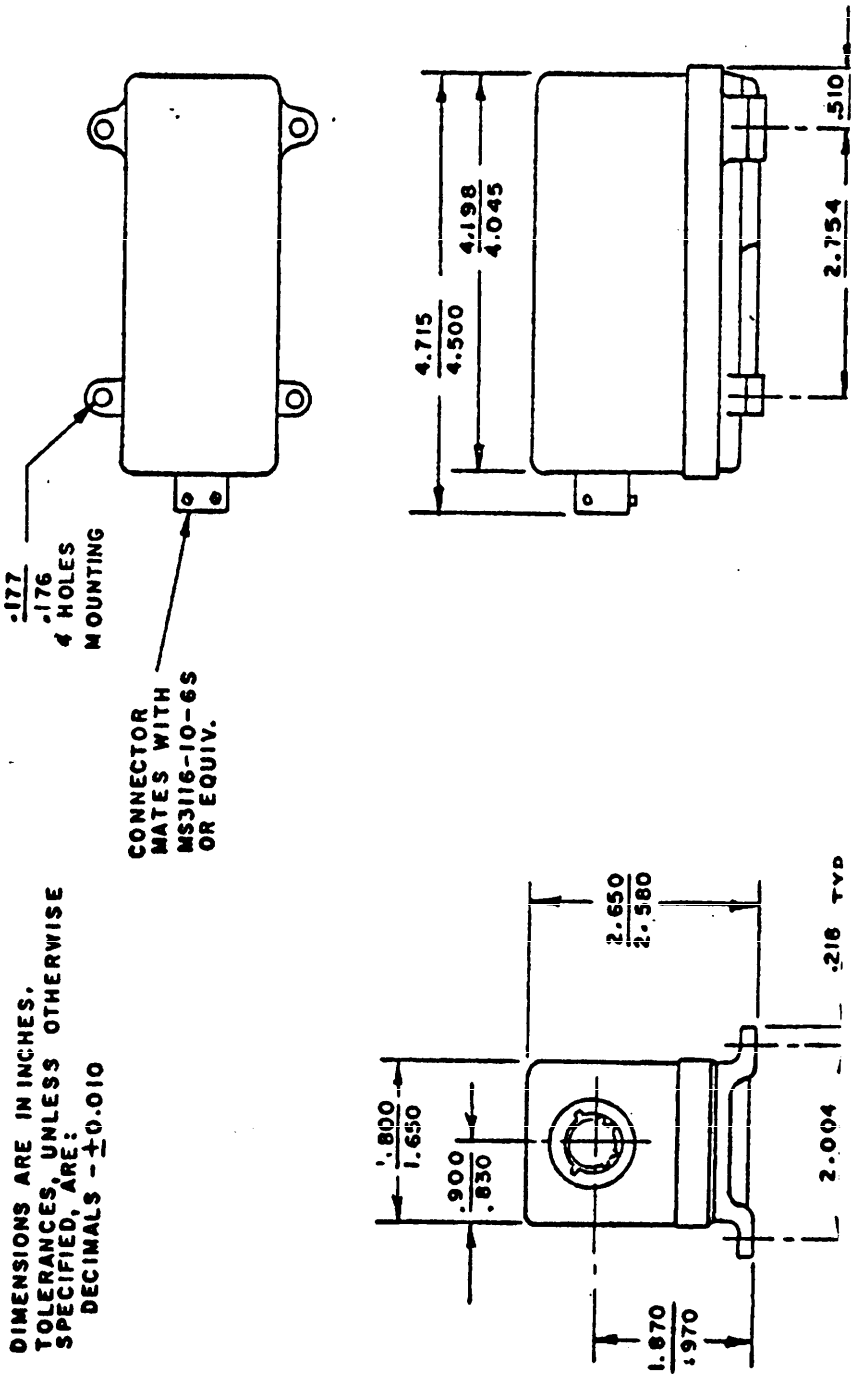


FIGURE 6 GYROSCOPE, RATE SWITCHING CN-495A/A

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4.2 First Article Tests - First article
sts shall be made on an equipment representative of each
oduction equipment to be supplied under the contract.
rst article tests shall be accomplished under the respon-
bility of the contractor and shall be conducted in accor-
nce with the approved test procedure of 4.6. The government
spector and the procuring activity shall be advised when
sts are to be conducted so that a representative may be
signated to witness or supervise the tests when so desired.
ntractors not having adequate facilities to conduct all
quired tests shall obtain the services of a commercial
sting laboratory acceptable to the Government.

4.2.1 First Article Test Data - The contractor
all submit all data collected in conducting these tests to
e procuring agency for review and approval. On first pro-
ction contracts this data shall include a list of all elec-
ical and electronic parts giving their specified voltage,
rrent and temperature rating and applied circuit voltage,
rrent and ambient and surface temperatures. The ambient
d surface temperature shall be obtained under the extreme
gh temperature operating condition.

4.2.2 Scope of Tests - First article tests
the individual equipments shall include all tests deemed
cessary by the procuring activity to determine that the
uipment meets all the requirements of this specification
d the contract. First article tests shall include environ-
ntal tests in accordance with the procedures of MIL-T-5422
d interference tests in accordance with MIL-I-6181.

4.2.3 First Article Approval - Approval of
e first article sample shall be by the procuring activity
on satisfactory completion of all tests. No production
uipments shall be delivered prior to the approval of the
rst article sample. Prefabrication of production equip-
nt prior to the approval of the first article sample is at
e contractor's own risk. The approved first article sample
all be retained by the contractor for his use in the fabri-
tion and testing of equipment to be submitted for accept-
ce. The first article sample shall not be considered as
e of the equipments under the contract.

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4.2.4 Production Equipments - Individual equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved first article sample. Each equipment shall be capable of successfully passing the same tests as imposed on the first article sample. Evidence of non-compliance with the above shall constitute cause of rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first 10 of the individual production equipments shall be selected and sent at the contractor's expense to a designate Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The first article sample shall not be selected for this test.

4.3.1 Scope of Tests - The individual equipments may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved first article sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tes the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production sample of each individua equipment shall be by the procuring activity upon satisfact completion of all tests. Any design, material or performan defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activit Failure of the Initial Production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approve and accomplished. Corrective action shall also be accompli on equipment previously accepted when requested by the procuring activity.

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4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

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4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient
Humidity	Ambient Room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum of 3 G's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the place for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directi the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken

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eliminate this defect from future equipment. A record ll be kept of all failures. The 6-hour period specified ve may be composed of two 3-hour periods to conform with ndard working hours.

4.4.2 Sampling Tests - Equipments selected sampling tests shall first have passed individual tests. ipments shall be selected for sampling tests by the governt inspector in accordance with the following: (Sampling ts shall not be conducted unless Reliability Assurance ts of 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	One for each additional 200 or fraction thereof.

TE: If by contract action the Initial Production Test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each ipment selected for sampling tests shall be subjected to following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the first article initial

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production, individual and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on t number of failures. However, each failure shall be analyzed as to cause and remedial action necessar to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Sampling Phase shall be conducted.

4.4.3.1 Reliability Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests." The maximum number of equipments to be used shall be those liste in Table 5 of MIL-STD-781. For the Qualification Phase, Tes Level E shall be used. The Accept-Reject Criteria for Test Plan III shall be used.

4.4.3.2 Reliability Sampling Phase Tests - Samples of the equipment shall be tested as outlined in MIL-STD-781, under the section entitled "Sampling Phase of Product on Reliability Tests." For the Sampling Phase Test Level E shall be used. The Accept-Reject Criteria for Test Plan V shall be used to determine the length of the tests (until an accept or rejection decision is reached).

4.4.3.2.1 Lot Size for Sampling Phase - The entire contract quantity (with the exception of those equipments used in the Qualification Phase) shall be considered one lot. The test program shall start the first month after the Qualification Phase has been completed. Samples shall be selected and placed on test in the same manner as though one month's production was a lot. The samples shall be tested until an accept or rejection decision is reached.

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: the meantime other equipments constructed shall be
ıipped. The test results shall be summarized monthly for
.e procuring activity. The procuring activity reserves
ıe right to stop the acceptance of equipment at any time
:ter one or more rejection decisions have been reached,
:nding a review of the contractor's efforts to improve the
quipment, the equipment parts, the equipment quality con-
'ol, etc., so that the entire contract quantity will show
ı accept decision.

4.4.3.3 Test Details - The test details such
: the length of the Test cycle, the length of the heat
ırtion of the cycle, the performance characteristics to be
ıasured, special failure criteria, preventive maintenance to
ı allowed during the test, etc. shall be part of the test
'ocedures to be submitted and approved by the procuring
:tivity prior to the beginning of the Qualification Test
ıase of the Reliability Assurance Tests.

4.4.4 Special Tests - Special tests shall be
ınducted on a quantity of equipments for the purpose of
ıecking the effect of any design or material change on the
:rformance of the equipment and to assure adequate quality
ıntrol. The equipment selected for special tests may be
:lected from equipments previously subjected to the sampling
· reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of
quipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall
ınsist of such tests as approved by the procuring activity.
:st procedures previously approved for the first article
:sts shall be used where applicable. When not applicable,
ıe contractor shall prepare a test procedure and submit it

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to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Longevity Tests - Longevity tests shall be performed in accordance with Test Plan XXVIII of MIL-STD-781 except that 500 hours operating time shall be performed on one sample of the equipment. One equipment used on the Reliability Qualification Plan may be continued until it has accumulated the required 500 hours. The longevity test procedures shall be similar to those of 4.4.3.

After the first 100 hours on the longevity test other maintenance may be performed provided the cost maximum of servicing and replacement of parts, throughout the longevity test, does not exceed the "cost to repair" specified in the contrac (see 6.4).

4.5.1 Reconditioning of Longevity Test Samples - An equipment which has been subjected to the longevity test shall be reconditioned as follows:

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(a) On completion of the longevity test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(b) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used or conducting first article tests, acceptance tests and ongevity tests shall be prepared by the contractor and sub- itted to the procuring activity for review and approval. The ight is reserved by the procuring activity or the government nspector to modify the tests or require any additional tests eemed necessary to determine compliance with the require- ents of this specification or the contract. MIL-T-18303 shall e used as a guide for preparation of test procedures. When pproved test procedures are available from previous contracts uch procedures will be provided and may be used when their se is approved by the procuring activity. However, the ight is reserved by the procuring activity to require modifi- ation of such procedures, including additional tests, when eemed necessary.

4.7 Reconditioning of Tested Equipment - quipment which has been subjected to acceptance and life ests shall be reconditioned by the contractor by replacing ll wear or damaged items. After reworking the contractor hall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part r complete equipment shall be submitted by the contractor ntil it has been previously tested and inspected by the ontractor and found to comply, to the best of his know- edge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which as been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection

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and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555.

6. NOTES

6.1 Intended Use - The AHRS defined by th specification is intended for use as an all-attitude headir and reference system for aircraft. The system provides rol pitch, and heading signals to the pilot's indicators and tc other avionic equipment.

6.2 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, a an improvement in the performance and reliability of the specific functions beyond the requirements of this specific tion are objectives which shall be considered in the produc tion of this equipment. Where it appears a substantial re- duction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability wil result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can b realized a request for approval shall be submitted to the procuring activity for consideration. Each request shall b accompanied by complete support information.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or appli- cable subsidiary specifications are in conflict, the follow ing precedence shall apply:

1. Contract - The contract shall have precedence over any specifi- cation.

2. This Specification - This specifi cation shall have precedence over all applicable subsidiary specifi cations. Any deviation from this

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specification, or from subsidiary specifications, where applicable shall be specifically approved in writing by the procuring activity.

3. Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications. Referenced specifications shall apply to the extent specified.

6.4 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

1. Title, number, and date of this specification

2. Selection of applicable levels of packaging and packing (see 5.1)

3. Cost to repair (see 4.5)

6.5 Associated Equipment - The AHRS shall operate with the following associated equipment which shall not be supplied.

Item	Type Designation
Transmitter, Remote Compass	ML-1

6.6 Performance Characteristics to be Measured -

Roll, Pitch, Heading Accuracy (At each 45° heading)	Repeat platform within ±1°
Dynamic Operation (About all three axes)	Hunting or jumping not to exceed 1°
Direction Gyro Drift	Not to exceed 3° per hour when corrected for earth's rate

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Heading Drift	Not to exceed 3° per hour with azimuth compensation but no correction for earth rate
Vertical Gyro Drift (with spin axis deflected 20°)	Not to exceed 25° per hour when corrected for earth's rate
Mode Operation	Synchronize within $\pm 3^{\circ}$

6.7 Definitions -

6.7.1 Hermetic Seal - A hermetic seal shall be a perfectly closed and airtight seal made between metall and metallic or between metallic and citric materials. A hermetic seal is not intended to include seals which are dependent upon gaskets.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-A-81423(AS) ATTITUDE HEADING REFERENCE SYSTEM AN/ASN-73

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

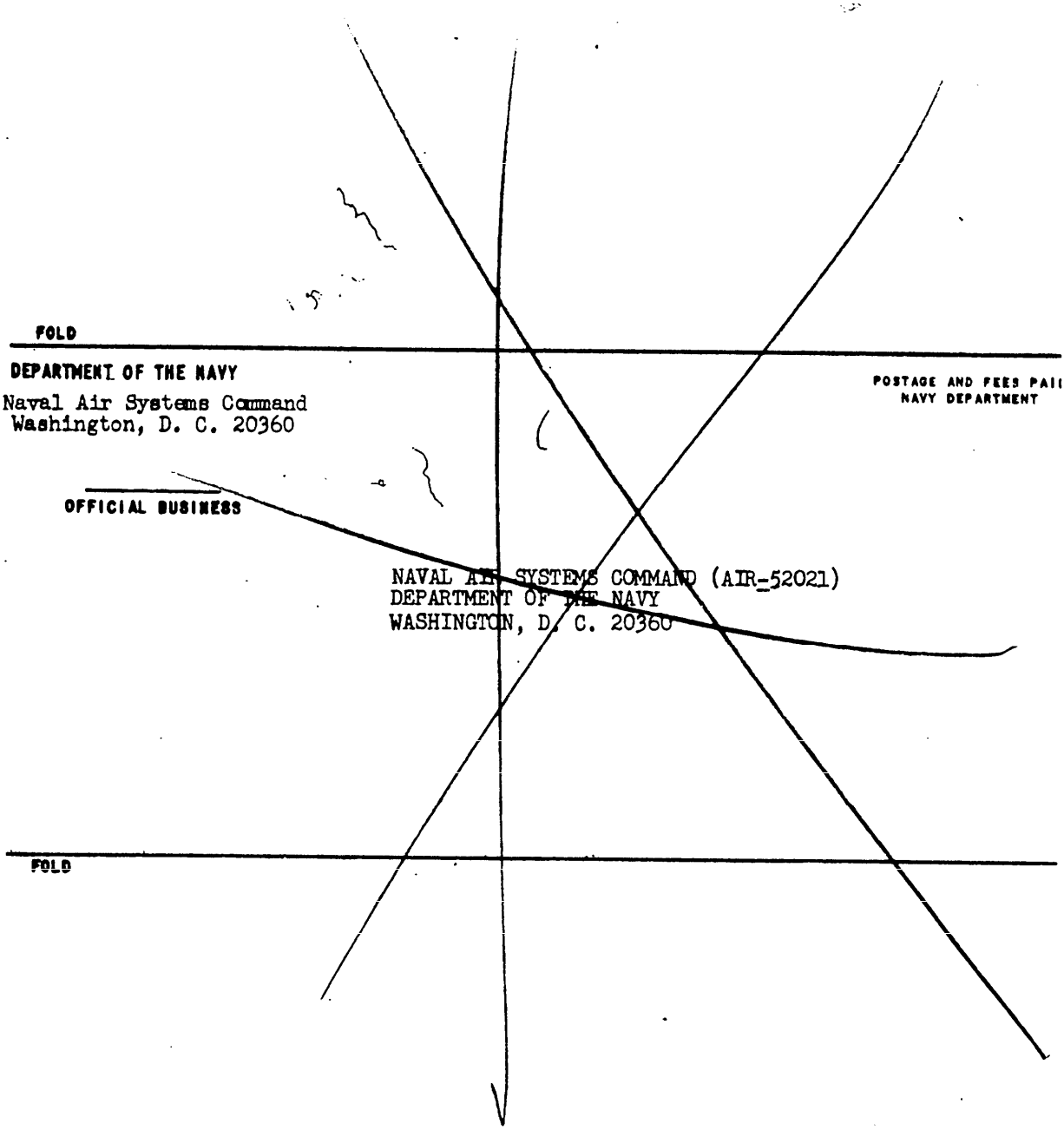
4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Naval Air Systems Command
Washington, D. C. 20360

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MIL-A-81423(AS)
Amendment -2

12 October 1972
Superseding
Amendment -1
28 September 1967

MILITARY SPECIFICATION

ATTITUDE HEADING REFERENCE SYSTEM AN/ASN-73

This amendment forms a part of Military Specification MIL-A-81423(AS) dated 1 October 1966 and has been approved by the Naval Air Systems Command, Department of the Navy.

Page 1

1.2: Delete "CN-105A/A" and "C-2984/AJB-3A" and substitute

"...CN-1054A/A..." and "C-2984/AJB..."

Page 2

2.1 Add the following under military specifications:

MIL-L-6085	Lubricating Oil, Instrument, Aircraft, Low Volatility
MIL-G-81322	Grease, Aircraft, General Purpose Wide Temperature Range
MIL-B-81744	Barrier Coating Solution and Solvent

Page 3

Add the following under military standards:

MIL-STD-1334	Process for Barrier Coating of Anti-Friction Bearings

Add the following under Naval Air Systems Command:

AR-34	Failure Classification for Reliability Testing, General Requirements for
AS-2844	Lubricant, Instrument Ball Bearing, High Flash Point

FSC 6610

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Page 7

3.3.2.4: Delete "MS17323" and substitute

"MS17322-8A"

Page 9

3.3.10.1: Delete and substitute

"3.3.10.1 Vibration - The equipment, less the Compass Controller, shall operate satisfactorily when subjected to the vibration requirements of Curve IV and II of Specification MIL-E-5400. The Compass Controller shall operate satisfactorily when subjected to the vibration requirements of Curve III of Specification MIL-E-5400 except that the maximum acceleration shall be limited to two (2) G's."

3.3.11, lines 3 and 4: Delete "...3.5 minutes .... and 5 minutes..." and substitute

"...5 minutes... and 15 minutes..."

Page 10

3.3.13.2, line 10: Delete "such that it will pass the case lead test." and substitute

"such that it will pass the case leak test."

Add the following new paragraphs:

* 3.3.16 Lubrication -

* 3.3.16.1 Rotor Bearing Lubrication - The rotor bearings shall be lubricated with grease conforming to MIL-G-81322 or MINIPURE grease with particle size not exceeding 50 microns.

* 3.3.16.2 Bearing Lubrication - The lubricating oil used in gimbal bearings shall be AS-2844 in accordance with the requirements of MIL-L-6085.

* 3.3.16.3 Barrier Film Bearing Coating - All bearings shall be processed in accordance with MIL-STD-1334. The coating material shall be in accordance with MIL-B-81744. All bearings, including chamfered races or corners, and including interfacing parts where applicable, shall be barrier coated.

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* 3.3.16.4 Ball Bearings - Ball bearings shall be ABEC-5 grade or better. Teflon separators shall be used on gimbal bearings where the bearing design permits. Spin rotor bearings shall have porous separators impregnated with lubricant AS-2844 and shall be ABEC-7 or better.

* 3.3.16.5 Ball Bearing Pre-Loading - Means shall be provided for pre-loading and alignment without causing eccentric loading unless otherwise authorized.

Page 11

3.4.3, line 3: Delete "the statistical accuracies of output signals" and substitute

"the statistical one sigma accuracies of output signals"

Page 18

3.5.1.1.5, paragraph 3, line 5: Delete "Table II" and substitute

"Table I"

3.5.1.1.5.1.1: Delete and substitute

"3.5.1.1.5.1.1 Externally Excited Pitch Synchro - Terminals c and K, Table I, shall be connected to the power lead terminal A, Table II, and terminal L of Table I shall be connected to the power lead terminal D, Table II. The voltage across terminals a and b, and the voltage across terminals L and a or b of Table I shall be measured with the gimbals at zero displacement. The voltage across L and a or b of Table I shall be less than the voltage applied to terminals A and D of Table II. The voltage across terminals a and c, and b and c, of Table I shall be measured as the reference assembly is rotated in climb about the pitch gimbal. The voltage across terminals a and c, Table I, shall increase before decreasing and the voltage across the terminals b and c, Table I, shall decrease before increasing."

Page 23

3.5.1.1.5.1.3: Delete and substitute

"3.5.1.1.5.1.3 Inner Roll Synchro - Terminal j of Table I shall be connected to terminal A of Table II. The voltage across g and h of Table I and the voltage across terminal D of Table II and terminal g or h of Table I shall be measured with the gimbals at zero displacement. The voltage across terminals g and h shall be at the minimum and the voltage across terminal D of Table II and terminals g or h of Table I shall be less than the voltage applied to terminals A and D of Table II. The voltage across terminals g and j and across h and j of Table I

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3.5.1.1.5.1.3 (Continued) shall be measured as the first gimbal (inner roll) is held and the reference assembly with the outer roll gimbal is rotated in right roll. The voltage across terminals g and j of Table I shall increase before decreasing and the voltage across terminals h and j shall decrease before increasing."

3.5.1.1.5.1.4: Delete and substitute

"3.5.1.1.5.1.4 Internally Excited Pitch Synchro - Terminal f of Table I shall be connected to terminal A of Table II. The voltage across d and e of Table I and the voltage across terminal D of Table II and terminal d or e of Table I shall be measured with the gimbals at zero displacement. The voltage across terminals d and e shall be at the minimum and the voltage across terminal D of Table II and terminal d or e of Table I shall be less than the voltage applied to terminals A and D of Table II. The voltage across terminals d and f and across e and f of Table I shall be measured as the second gimbal (pitch) is held and the reference assembly with theouter roll gimbal is rotated in climb. The voltage across terminals d and f of Table I shall increase before decreasing and the voltage across terminals e and f shall decrease before increasing."

Page 24

3.5.1.1.5.1.5, line 7: Delete "Table I" and substitute

"Table II"

Page 25

3.5.1.1.9, lines 4 and 7: Delete "±90" and substitute

"+90"

Page 26

3.5.1.2.4.1, line 7: Add at the end of the sentence

"The synchro shall not exceed a maximum transmission error of 8 minutes of arc."

Page 27

3.5.1.2.4.3, lines 9 and 18: Delete "Table II" and substitute

"Table I"

3.5.1.2.4.3, line 10: Delete "Table I" and substitute

"Table II"

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3.5.1.2.4.3.1, line 2: Delete "Table I" and substitute

"Table II"

Page 29

3.5.1.3, line 3: Add page "30."

Page 40

Table VIII: Delete "To mate with MS3116P-10-6S or equal (5 req'd)" and substitute

"To mate with MS3116P-10-6S or equal (4 req'd)"

Table VIII: Delete

Terminal	Function
D	CX Excitation C
E	CX Excitation H

and substitute

Terminal	Function
D	CX Excitation H
E	CX Excitation C

Page 42

Table X: Add

Terminal	Function
G	+28 vdc rectified
J	Spare
K	Spare

Page 43

Table XI: Delete "Pre-Amplifier-Limited Output" and substitute

"Pre-Amplifier-Limiter Output"

Table XI: Delete "Slaving and Correction Motor" and substitute

"Slaving and Correction Velocity Generator"

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Page 45

3.5.3.4, line 5: Delete "Compensator shall increase (+) when the" and substitute

"Compensator shall increase (+) when the"

3.5.3.5, line 8: Delete "Items a, b, d, and e shall be" and substitute

"Items a, b, and d shall be"

Page 46

3.5.3.7, paragraph 2, line 1: Delete "With the CX (J908) at" and substitute

"With the CX (J906) at"

3.5.3.7, paragraph 2, lines 2 and 3: Delete "CX's (J906, J907 and J909)" and substitute

"CX's (J908, J907 and J909)"

Page 47

3.5.3.9.4: Delete and Substitute

"3.5.3.9.4 Synchronization Indicator - The null current through the 1100 ohm resistor shall not exceed 8 microamperes. The current through the 1100 ohm resistor, throughout a 5 degree displacement of the magnetic comparison transmitter shall be 10 ± 4 microamperes per degree"

3.5.3.10: Delete "(2) 23.5V, 400 cps for excitation of the type MX-1 magnetic compass transmitter." and substitute

"(2) 23.5V 400 cps for excitation of the type ML-1 magnetic compass transmitter."

Page 48

3.5.4: Delete "Controller-Compass System C-2984/AJB-3A" and substitute

"Controller-Compass System C-2984/AJB"

Page 49

Figure 5: Delete title and substitute

"Controller Compass System C-2984/AJB"

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Page 58

4.4.1.3 Delete (Under Vibration) "Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and minimum of 3G's." and substitute "any selected frequency within the range of 50 to 60 cps (excluding resonant points) and a minimum of 3G's."

Page 60

4.4.3 Add the following to end of paragraph:

* Classification of a failure shall be in accordance with MIL-STD-781 and AR-34.

Add the following paragraph:

* 4.4.3.1.1 Reliability Assurance Production Acceptance Phase - Reliability Assurance Production Acceptance Testing shall be conducted through a production burn-in. Each equipment produced but not submitted to the Qualification Test shall be operated for 150 operating hours burn-in, and the test data (operating hours and failures) shall be totaled and compared to the Accept/Reject Criteria of Test Plan II of MIL-STD-781. (Extend the lines as necessary to accomodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. Delivery of equipment shall not be held up to await a decision under the test plan. In event an Accept Decision is reached, the test duration of each equipment shall be reduced to 100 operating hours. At any time that the current total of test hours and test failures plotted on Test Plan IV curves (extended) show a Reject Situation, the Procuring Activity shall be notified. The Procuring Activity reserves the right to stop the acceptance of equipment at any time that a Reject Situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a rejection decision. For the Production Acceptance Phase, Test Level E shall be used, except that the temperature extremes shall be +55C and -44C, and vibration shall be 0.5G. Details of the burn-in tests shall be included in the test procedures.

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Preparing Activity:
Navy - AS

Project No. 6610-N065

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